Computational Complexity and Energetics of the Ant Stack

Analytical Workload Models and Explicit Energy Accounting

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1 Abstract

We present an analytical framework for studying computational complexity and energy accounting in the Ant Stack, a proposed Body–Brain–Mind architecture for embodied artificial intelligence. Closed-form workload counters connect contact-resolution, sparse neural-processing, and policy-evaluation models to explicitly assumed energy coefficients. A nominal 100 Hz update rate defines a 10 ms accounting interval; it is not a measured scheduling guarantee.

The selected manifest uses the base workload family: contact models named `pgs`, `si`, and `ldlt`; neural patterns named `random` and `structured`; and an exact policy-tree model parameterized by branching factor B and horizon H_p in decision steps. The separately available enhanced Mind model bounds the policy count by `max_policies` (default 1000), with `None` selecting exact enumeration. These are different analytical models, not interchangeable evidence of bounded-policy performance.

Energy is estimated from operation counts, memory traffic, spikes, and baseline power, with mechanical and sensing terms reported on an explicitly matched time basis. The coefficients are illustrative assumptions, not measurements calibrated to a named device. Current numerical tables and fitted relationships belong in the run-generated `Generated.md`, rather than duplicated constants in this prose. Bootstrap resampling describes the supplied model values; repeated deterministic calculations do not establish physical uncertainty or statistical power.

The contribution is a transparent model and reproducible analysis interface, not empirical validation of ant physiology, hardware prediction accuracy, real-time feasibility, critical exponents, or species-specific behavior. Independent measurements and a declared validation protocol are required before making those claims. The model provides a basis for asking how solver choice, neural activity, and policy budgets affect an assumed computational energy budget, while keeping implementation checks distinct from biological evidence.

2 Computational Complexity and Energetics of the Ant Stack

2.1 Analytical Scope and Evidence Boundary

This companion manuscript studies **analytical workload and energy models**, not measured validation of an ant, robot, or processor. Body, Brain, and Mind are conceptual accounting boundaries. Input coefficients and dimensions are illustrative assumptions; current numeric results belong only in `Generated.md` from the corresponding completed governed run.

Compute-aware planning research (Sudhakar et al. 2020) motivates comparing computation and actuation costs. Historical computing-efficiency trends (Koomey et al. 2011) do not calibrate this example's coefficients. No independent raw calibration, species-validation dataset, or demonstrated publication-readiness result is identified here.

2.2 Reproducible Inputs and Products

`manifest.example.yaml` selects base closed-form workloads, supported `pgs`/`si`/`ldlt` contact models and `random`/`structured` neural patterns, canonical Mind B, H_p, and a null meter. Workload rows

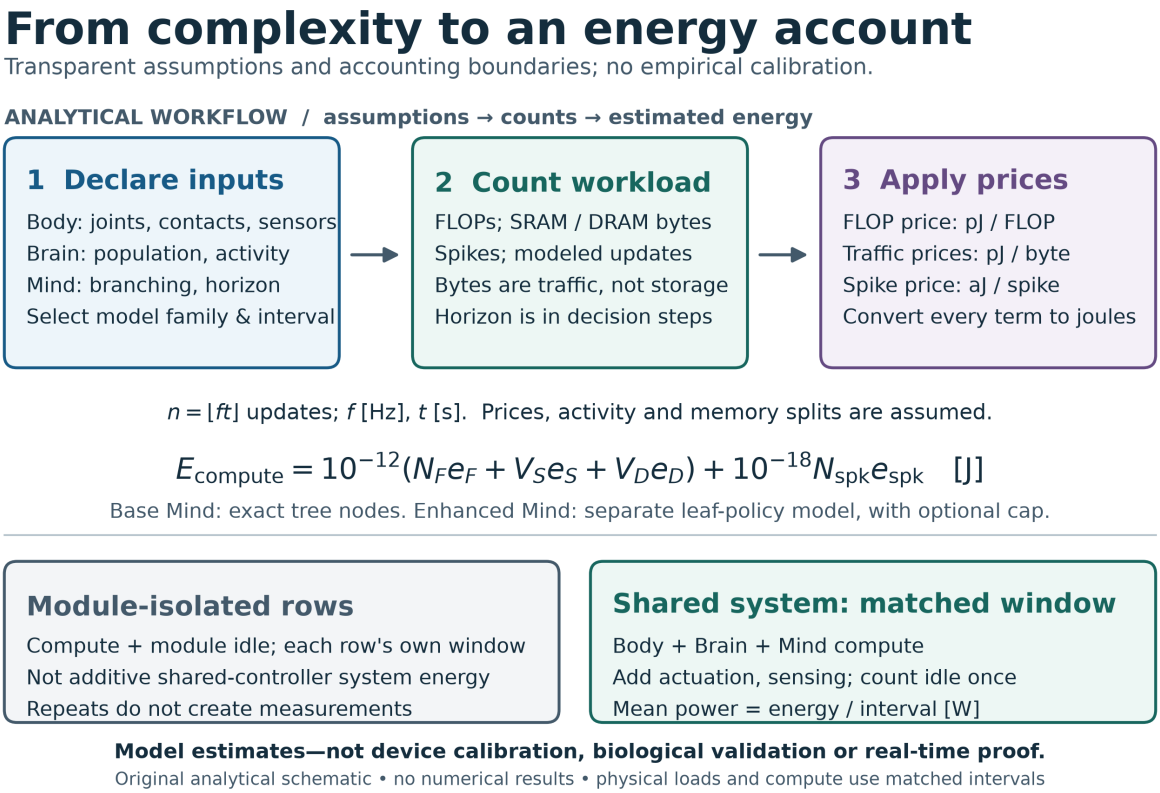


Figure 1: From Workload Assumptions to Matched Energy Accounting: Original analytical schematic: assumed dimensions and energy prices produce model estimates, not empirical calibration. Module-isolated rows and common-window system components have different accounting boundaries. Source and output identities accompany the graphical assets.

are isolated compute-plus-idle estimates over their windows. The separate system per-decision table uses the same base compute models, assumed physical actuation/sensing, and one shared idle baseline. Mind physical compute is not zero. CoT is unavailable without a matched modeled locomotion interval for the distance.

The package CLI writes every CE product beneath the fresh normalized `--out` destination: tabular/provenance outputs, `assets/`, and `Generated.md`. Governed `run-all-antstack` publishes canonical results beneath `outputs/<run_id>/` and projects explicitly completed products to paper-local compatibility `out/`, `assets/`, and `Generated.md`, preserving source-owned guides. Do not hand-copy arbitrary output-tree contents or treat a historical compatibility artifact as a completed fresh run.

From the repository root, the release owner can generate a fresh CE result tree (destination must not already contain a prior run):

```
uv run antstack-ce papers/complexity_energetics/manifest.example.yaml --out
outputs/ce-manuscript-fresh
```

For governed canonical orchestration and compatibility projection, use the root guide's `run-all-antstack` configuration and command. Publication configuration marks CE analysis as required and uses the package module, not the removed paper-local runner. Rendering is a separate governed acceptance step. No generation or render is claimed by these source edits.

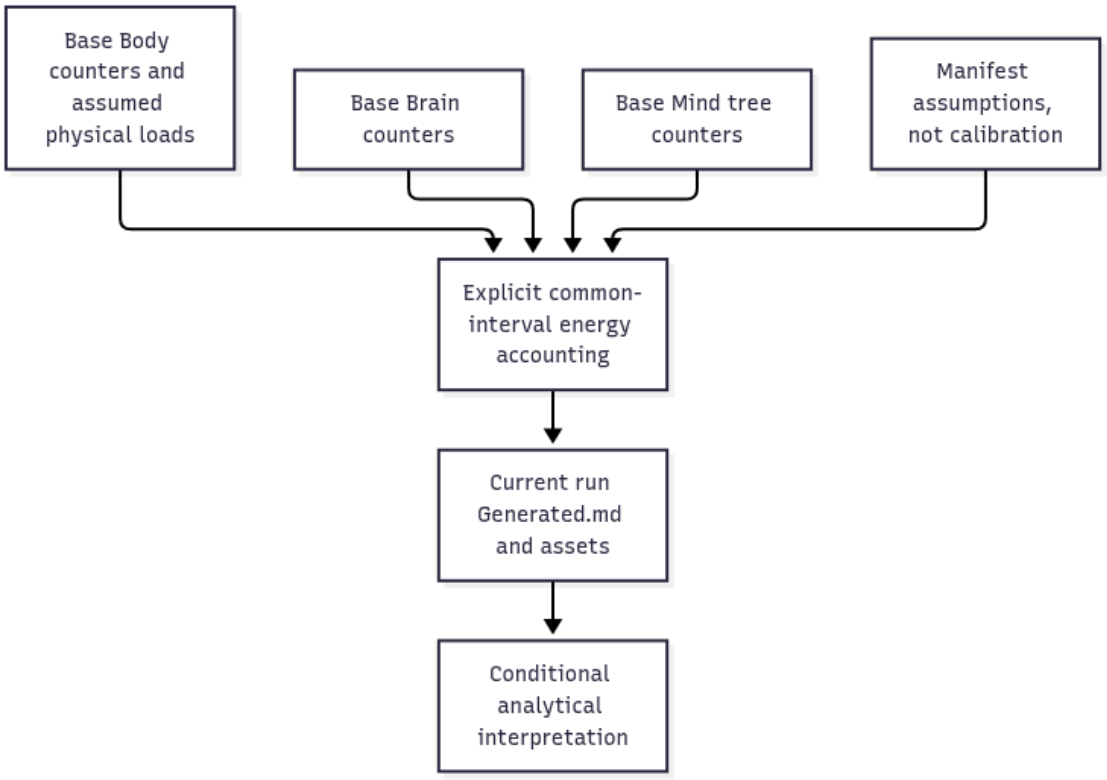


Figure 2: Ant Stack Overview and Flow: Analytical module counters and assumed physical loads feed matched-interval accounting

Caption: Source-owned conceptual schematic. Numeric outputs are generated, while coefficients remain assumptions until independently calibrated. Separate enhanced models are not implicitly substituted for selected base workloads.

3 Background and Research Context

3.1 Research Motivation and Scope

Embodied systems connect computation, sensing, and physical action. Their energy budgets depend on algorithmic work, memory traffic, hardware operating conditions, and mechanical tasks. The Ant Stack provides a conceptual Body–Brain–Mind decomposition for asking how these contributions interact. This companion study analyzes closed-form workload models; it does not report a calibrated robot or validated ant emulation.

The central questions are which configured dimensions drive modeled work, how assumed operation and power prices affect the energy budget, and which additional measurements would discriminate competing design choices. A nominal 100 Hz update target is not an observed control capability. Neither contact count 20 nor planning horizon 15 is a hardware-independent feasibility boundary.

3.2 Energy-Aware Robotics and Computational Co-Design

Platform-specific robot models must specify motion, payload, and hardware. Jaramillo-Morales and colleagues model differential-drive mobile robots and validate on their identified platform (Jaramillo-Morales et al. 2020); this supports the importance of platform-specific validation, not an accuracy claim for this framework.

Compute Energy Included Motion Planning (CEIMP) explicitly weighs further planning energy against potential actuation savings (Sudhakar et al. 2020). It motivates treating compute and actuation together. The present CE sweep counts work and does not itself implement or evaluate CEIMP's stopping decisions.

Historical improvements in computing's electrical efficiency (Koomey et al. 2011) provide context for energy-aware design. They do not calibrate our FLOP, memory, or spike coefficients. Device-level or sparse-processing advantages remain workload-, technology-, and boundary-dependent; a generic attojoule spike price is an assumption here.

3.3 Architectural Context

AntBody represents morphology, contacts, sensing, and actuation. The analysis counts hypothetical contact/control/sensor work for configured dimensions, not a full biomechanical simulation or empirical locomotion trajectory.

AntBrain abstracts sensory and sparse neural processing inspired by insect AL, MB, and CX organization. Supported counter patterns are `random` and `structured`, not validated species topologies. Population size and activity fraction are illustrative input choices, not inferred biological measurements.

AntMind represents policy evaluation and active-inference concepts (Friston et al. 2015). The base counter model enumerates tree nodes; a distinct enhanced model bounds leaf policies. An expected-free-energy score is a decision-theoretic quantity, not automatically thermodynamic energy or a demonstrated biological mechanism.

Stigmergic coordination through proposed diffusion–decay fields is architectural context. The native CE manifest's module sweeps do not establish colony-size scaling, collective-task performance, or nonlinear energy coupling.

3.4 Theoretical Foundations and Applicability

Shannon's information theory (Shannon 1948) motivates explicit channel assumptions. For an ideal bandwidth-limited additive Gaussian-noise channel,

$$C = W \log_2(1 + \mathrm{SNR}) \quad \text{bits/s}. \tag{1}$$

Here W is bandwidth, not the planning branching factor B. Sensor-channel count alone supplies neither bandwidth, signal-to-noise ratio, nor achieved information rate. No percentage of channel capacity is inferred for AntBrain.

Landauer's principle concerns logically irreversible bit erasure under thermodynamic assumptions (Landauer 1961):

$$E_{\mathrm{erase}} \geq k_B T \ln 2 \quad \text{per erased bit.} \tag{2}$$

At approximately room temperature this reference is of order 10^{-21} J/bit. It is not an exact per-FLOP price, nor a complete workload minimum without a justified erasure count. Mechanical-work and information references require compatible boundaries before comparisons.

The heat-engine Carnot expression

$$\eta_{\mathrm{Carnot}} = 1 - T_c/T_h \tag{3}$$

applies between specified thermal reservoirs. It is not a generic electromechanical actuator-efficiency bound. Consequently this study does not claim a universal thermodynamic optimization factor from its assumed electrical prices.

3.5 Levels of Analysis and Evidence Gaps

At the **problem level**, proposed outcomes include navigation, foraging, task allocation, and robustness. At the **algorithm level**, representations and policy/solver choices determine hypothetical work. At the **physical implementation level**, devices, sensing rails, actuation, latency, and environmental conditions determine measurable resource use. Keeping these levels separate prevents operation-count consistency from being mistaken for successful behavior or calibrated physical predictions.

The workflow supplies an explicit input/result interface and model-based comparisons. Independent device microbenchmarks, held-out energy traces, identified biological datasets, and task outcomes are still required for accuracy, real-time, species, or biological-superiority claims. No current source-based result establishes Sobol/Morris sensitivity, critical phenomena, or external replication.

4 Analytical Operation-Count and Memory-Traffic Models

4.1 Workload Assumptions and Unverified Timing Targets

We analyze operation-count and traffic models, not executed contact solvers or biological neural circuits. Asymptotic time, modeled byte traffic, and energy estimates are distinct from wall-clock latency and peak storage. A nominal 100 Hz decision rate implies a 10 ms scheduling target; meeting it requires profiling and deadline evidence on named hardware.

Parameterized formulas are useful for identifying which dimensions drive work. They do not establish a formal fixed-parameter-tractability classification, solver accuracy hierarchy, or hardware-independent feasibility boundary. In particular, neither 20 contacts, activity fraction 0.02, nor horizon 15 is a measured critical point.

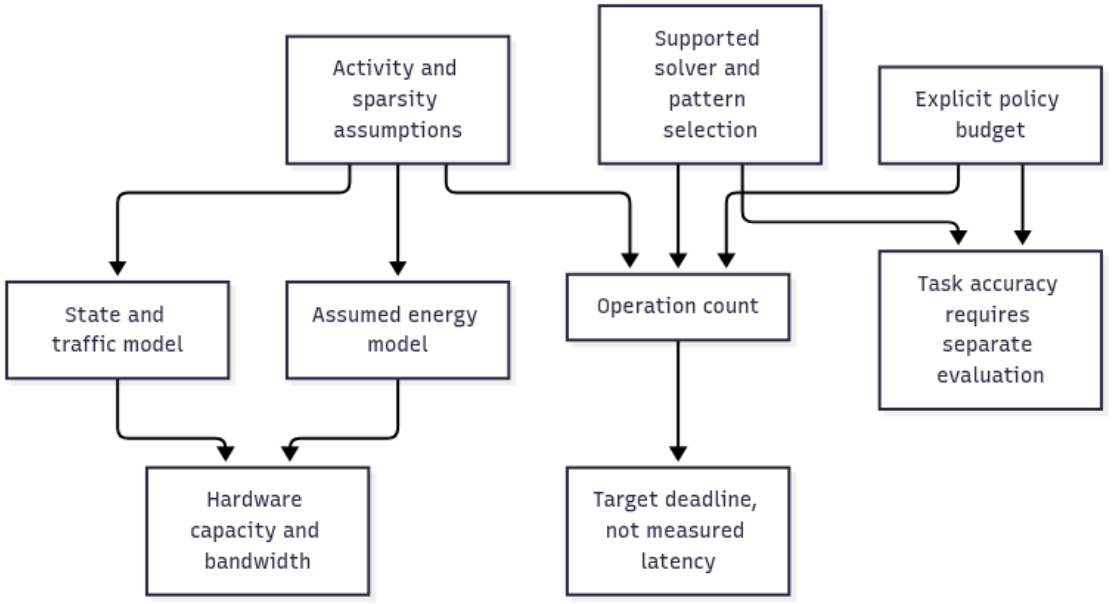


Figure 3: Complexity Trade-offs in Embodied AI: Conditional workload trade-offs; deadlines and hardware limits are not measured outcomes

Caption: Conceptual trade-offs between modeled work, hardware constraints, and unmeasured task quality. Algorithm and budget choices do not by themselves demonstrate accuracy or real-time operation.

4.2 System Parameter Framework

J denotes actuated joint degrees of freedom, C modeled contacts per update, S sensor channels, K neural input channels, N_{KC} population size, ρ active fraction, and H channel/heading-state dimension. B is the canonical planning branching factor and H_p the horizon in **steps**, not seconds. State/action dimensions and precision steps affect the policy counters.

The example dimensions and sweep ranges are engineering assumptions in the manifest, not species estimates or validated operating ranges. An 18-joint example corresponds arithmetically to six legs with three modeled joints each; it does not identify a biological ant's anatomy or mass. Neural-population and activity choices likewise are not calibrated ant measurements.

4.3 Selected Base Workloads

The CE orchestration selects the base closed-form functions. Each multiplies per-update counters by $n = \lfloor ft \rfloor$. Its modeled memory bytes are accumulated traffic over these updates; do not label them peak resident memory without a separate storage analysis.

4.3.1 AntBody Contact and Control Models

Only `pgs`, `si`, and `ldlt` are supported; unsupported solver names are rejected rather than silently aliased. The following are literal analytical cost conventions, not measured convergence or accuracy results.

Table 1: Analytical Contact-Cost Conventions

Name	Contact FLOPs per update	Contact memory convention (bytes)
`pgs`	$IC(6J+12)$, $I=\min(50,\max(10,C))$	$8(J^2+6C)$
`si`	$20CJ+J^3/3$	$8J^2+96C$
`ldlt`	$(J+3C)^3/3$	$8(J+3C)^2$

Base Body adds $10S$ sensor FLOPs, $20J^2$ control FLOPs, $8S$ sensor bytes, and $8J^2$ control bytes. Its traffic split is an assumed 80% SRAM and 20% DRAM. At fixed J, PGS contact work is piecewise linear/quadratic/linear as the capped iteration rule changes; it is not universally $C^{1.5}$. There is no solver-switching rule or tenfold iteration jump at $C=20$.

4.3.2 AntBrain Sparse-Population Model

For $N_a = \lfloor \rho N_{KC} \rfloor$, `random` uses N_a^2 neural FLOPs and $100N_a$ connections; `structured` uses $N_a \log N_{KC}$ FLOPs and $50N_a$ connections. The shared memory convention is four bytes per connection and population state, and the spike-event convention is $0.1N_a$ per update. These are assumed counters, not sampled biological spikes.

Base Brain adds $100K$ channel FLOPs and $4KH$ channel bytes, then assumes a 90% SRAM/10% DRAM traffic split. Therefore increasing K at fixed other parameters is not exactly free. No small-world, scale-free, or biological topology is implemented under those names, and no measured biological efficiency ranking is implied by choosing `structured`.

4.3.3 AntMind Exact Tree Model

Base Mind uses

$$N_{\text{tree}} = \sum_{h=0}^{H_p} B^h, \tag{4}$$

with belief work $10N_{\text{tree}}d_s^2$, policy work $5N_{\text{tree}}d_a q$, and evidence-bound work $3N_{\text{tree}}(d_s+d_a)$, where d_s, d_a, q are state dimension, action dimension, and precision steps. Its storage/traffic convention includes node states, transition matrices, and policy parameters, with an assumed 70% SRAM/30% DRAM split.

For fixed dimensions and $B>1$, exact tree enumeration grows exponentially with H_p. For $B=1$ the node count is H_p+1. These mathematical growth statements do not predict a universal deadline failure at a particular horizon. The functions count hypothetical work; they do not establish the efficacy of active-inference actions (Friston et al. 2015).

4.4 Separate Enhanced Workloads

Enhanced Body uses detailed optical-flow and sensor counters, $25J$ dynamics work, and quadratic control work. Enhanced Brain uses separate AL, MB, CX and plasticity terms; its CX convention includes $12H + H^2/2$ FLOPs. Neither description should be substituted for a base-model result.

Enhanced Mind uses $P = \min(B^{H_p}, M)$ for finite `max_policies=M`, uniformly across horizons; default $M=1000$, while `None` selects exact B^{H_p} leaf policies. Its belief/EFE/precision counters scale with P, with an optional assumed hierarchical overhead. A bound on policy count can saturate the

modeled work at fixed dimensions; it does not prove unchanged decision quality, guarantee real-time performance, or reduce the exact tree model to $\mathcal{O}(BH_p)$.

4.5 Environmental Fields and Cross-Module Coupling

A proposed explicit two-dimensional diffusion–decay field with G cells and E deposits costs $\mathcal{O}(G+E)$ for fixed-radius stencils and uses $\mathcal{O}(G)$ storage. For pure diffusion on a square grid of spacing h and diffusion constant $D > 0$, the explicit stability condition is $\Delta t \leq h^2/(4D)$; decay and boundary conditions require their own treatment. This field is architectural context, not an additional measured CE workload.

Body observations can feed Brain, Brain states can feed policy selection, and Mind decisions can modulate Body/Brain parameters. The selected workload sum does not estimate a nonlinear coupling matrix. A proposed expression such as

$$E_{\text{interaction}} = \sum_{i,j} \alpha_{ij} E_i E_j + \sum_{i,j,k} \beta_{ijk} E_i E_j E_k \tag{5}$$

would require justified coefficients with units J^{-1} and J^{-2} and independently identified responses. No fitted interaction energy or critical manifold is reported here.

Caption: Conceptual information flow with explicit intended sensor, channel, branching/horizon, and inference-dimension edges. The dashed edge distinguishes an alternative enhanced model, not a second additive processing stage. Labels are plain diagram text; numeric counters belong in current generated results.

Table 2: Parameter Meanings and Input Authority

Symbol	Meaning	Input authority
J, C, S	Joints, contacts, sensor channels	Body manifest parameters
K, N_{KC}, ρ, H	Channels, population, activity fraction, state dimension	Brain manifest parameters
B, H_p	Branching factor, horizon in steps	Mind manifest parameters
d_s, d_a, q	State/action dimensions, precision steps	Selected Mind model
G, E	Field cells, deposits per update	Proposed field protocol, not CE results

All numerical ranges remain assumptions unless bound to a cited dataset or actual measurement. Current fitted scaling and energy estimates are generated once in `Generated.md`, with their selected model family and accounting interval.

5 Analytical Energy Accounting and Assumed Coefficients

5.1 Accounting Boundary and Physical Energy Balance

This section defines analytical accounting, not measured platform performance. Digital prices, sensor power, per-joint actuator power, and idle power are illustrative assumptions in `manifest.example.yaml`; none is calibrated to a named processor, spike circuit, robot, or ant. There

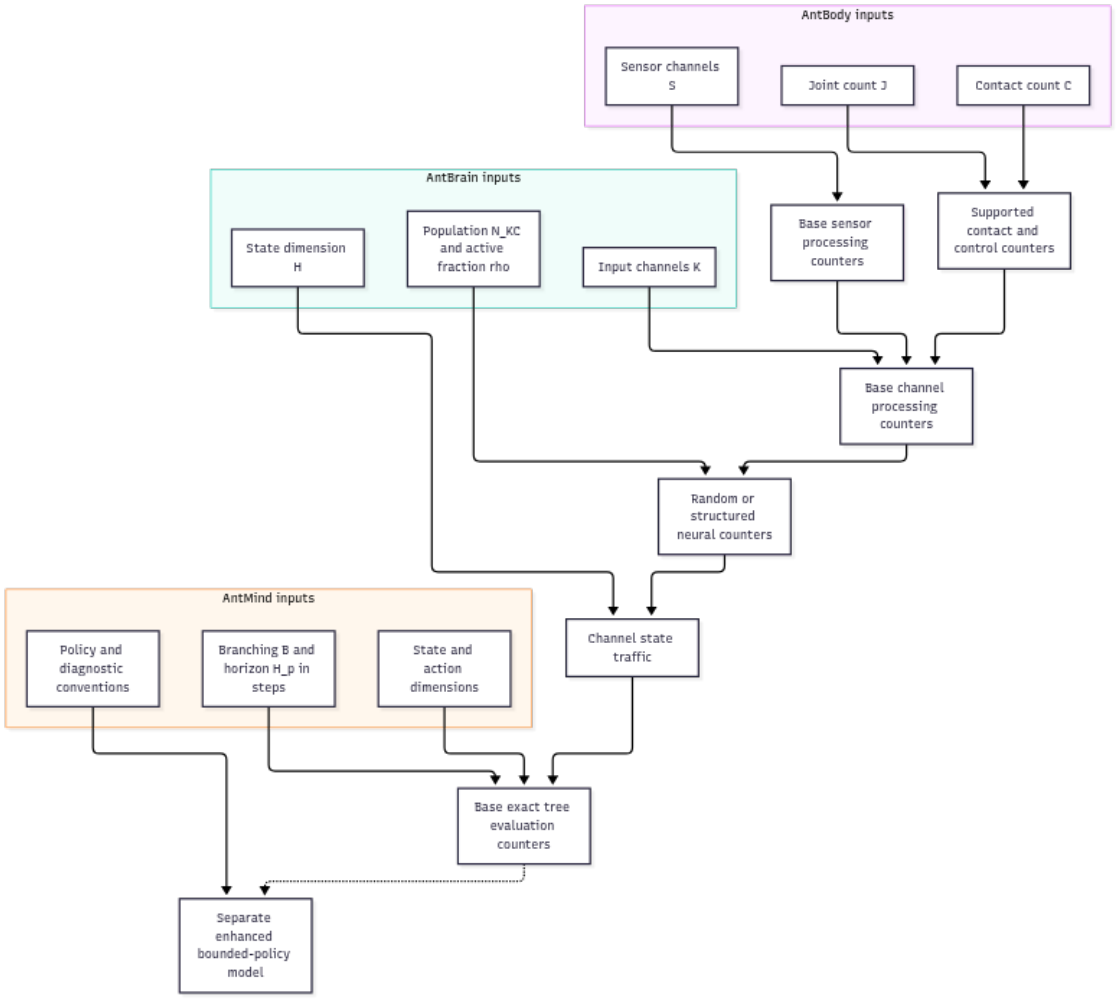


Figure 4: Module Complexity Overview: Declared dimensions drive separate Body, Brain and Mind analytical counters

is no demonstrated prediction error below 5%, 7 nm attojoule measurement, or species-specific efficiency estimate.

For an identified system boundary, the first-law balance is

$$\frac{dE_{\text{stored}}}{dt} = P_{\text{input}} - P_{\text{dissipated}} - P_{\text{work}}. \tag{6}$$

Here P_{work} is mechanical work leaving the boundary. Electrical compute consumption and its eventual heat are not separate additive electrical loads. Kinetic energy $mv^2/2 + I\omega^2/2$ and potential energy mgh are storage terms; operation-price accounting estimates an electrical input requirement, not a distinct conserved reservoir.

5.2 Compute, Memory, and Idle Energy

Over a window t, modeled digital energy is

$$E_{\text{compute}} = N_{\text{FLOP}} e_{\text{FLOP}} + V_{\text{SRAM}} e_{\text{SRAM}} + V_{\text{DRAM}} e_{\text{DRAM}} + N_{\text{spk}} E_{\text{spk}}. \tag{7}$$

Coefficients must first be converted to joules per operation, byte, or spike. Window baseline is $E_0 = P_0 t$. Byte counters denote modeled traffic, not measured rail energy or automatically peak storage. Spike prices omit unspecified host, routing, and I/O costs unless explicitly included in the coefficient boundary.

At the illustrative manifest baseline of 0.50 W, idle energy is 5 mJ per 10 ms, not 0.5 mJ. The 0.5 s workload window contains 50 updates at the assumed 100 Hz and 0.25 J idle energy. These dimensional consequences are not calibration results. Current aggregate numbers are generated in `Generated.md` and not copied into diagrams or prose.

5.3 AntBody: Mechanics and Sensing

For a specified torque/velocity trace, mechanical work is

$$W_{\text{mech}} = \int_{t_0}^{t_1} \tau(t)\omega(t)\, dt. \tag{8}$$

The corresponding electrical requirement depends on drive losses, regeneration, and which positive/negative work convention is used. For nonnegative delivered work and a specified efficiency,

$$E_{\text{elec}} = W_{\text{mech}}/\eta_{\text{drv}} + E_{\text{additional losses}}. \tag{9}$$

Do not add losses already absorbed into η_{drv} a second time. The current example instead uses an assumed per-joint electrical-power coefficient and an assumed sensor-power-per-channel coefficient; it is not a torque-trace measurement. Integrate both over the same interval as compute and idle before forming a complete system budget.

Terrain/friction, duty cycling, and sensor selection can change power. No fixed terrain penalty of 20–40%, duty-cycle saving of 50–80%, universal actuator efficiency, muscle/motor power-density ranking, or battery-to-carbohydrate improvement ratio is established here. Such comparisons require specified devices, operating conditions, and primary measurements. Robot-specific power modeling (Jaramillo-Morales et al. 2020) illustrates why those conditions matter.

5.4 AntBrain: Sparse-Processing Assumptions

The selected base neural workload combines the supported `random` or `structured` population model with channel processing. Enhanced AL–MB–CX counters are a separate model, not the producer of base sweep rows. Activity fraction ρ, memory splits, and spike-event counts are modeling conventions. No optimal biological activity fraction, small-world comparison, or measured event-driven energy-saving factor is inferred from them.

Historical processor efficiency trends (Koomey et al. 2011) motivate attention to energy, but do not set these device prices. Use independently calibrated coefficients before making hardware predictions. Circuit-only spike energy must not be compared with a complete software-system energy without reconciling their boundaries.

5.5 AntMind: Conceptual Layer and Physical Compute

Mind is a conceptual policy/semantic layer, not a proof of zero physical energy. Policy evaluation runs on physical hardware and its base-model FLOP/memory counters contribute to the compute budget. If that work is assigned to a Brain processor, reassign the accounting boundary rather than dropping or double-counting it.

The base Mind model enumerates tree nodes through H_p steps using B. The separate enhanced model caps leaf-policy count with `max_policies` (default 1000), or uses exact enumeration for `None`. Neither model establishes a universal horizon of 15 steps or an energy-free symbolic implementation. A time horizon in seconds requires an explicit planning-step duration.

For utilization-based profiling on a particular device, one possible model is

$$E_{\text{mind}} = [uP_{\text{active}} + (1-u)P_{\text{idle}}]\Delta t. \tag{10}$$

This is an alternative accounting model requiring measured/utilization inputs, not an extra term to add on top of already priced operations without reconciling overlap. Active inference's variational or expected free energy (Friston et al. 2015) is not automatically thermodynamic energy in joules.

5.6 Integrated Per-Decision Accounting

For a common decision interval, the complete modeled sum is

$$E_{\text{decision}} = E_{\text{actuation}} + E_{\text{sensing}} + E_{\text{body compute}} + E_{\text{brain compute}} + E_{\text{mind compute}} + E_{\text{shared idle}}. \tag{11}$$

The generated per-decision table uses the selected base compute functions and the configured update rate, with assumed physical terms and one shared idle load. In contrast, raw workload rows are module-isolated compute-plus-idle energies over their individual windows. They are not directly additive system components. Repeats are not additional simultaneous modules or independent measured traversals.

Caption: Analytical accounting boundaries, not measured energy flows. All summed terms share one interval; Mind physical compute and Body compute are retained. Numerical values are generated from the current manifest rather than fixed in this schematic.

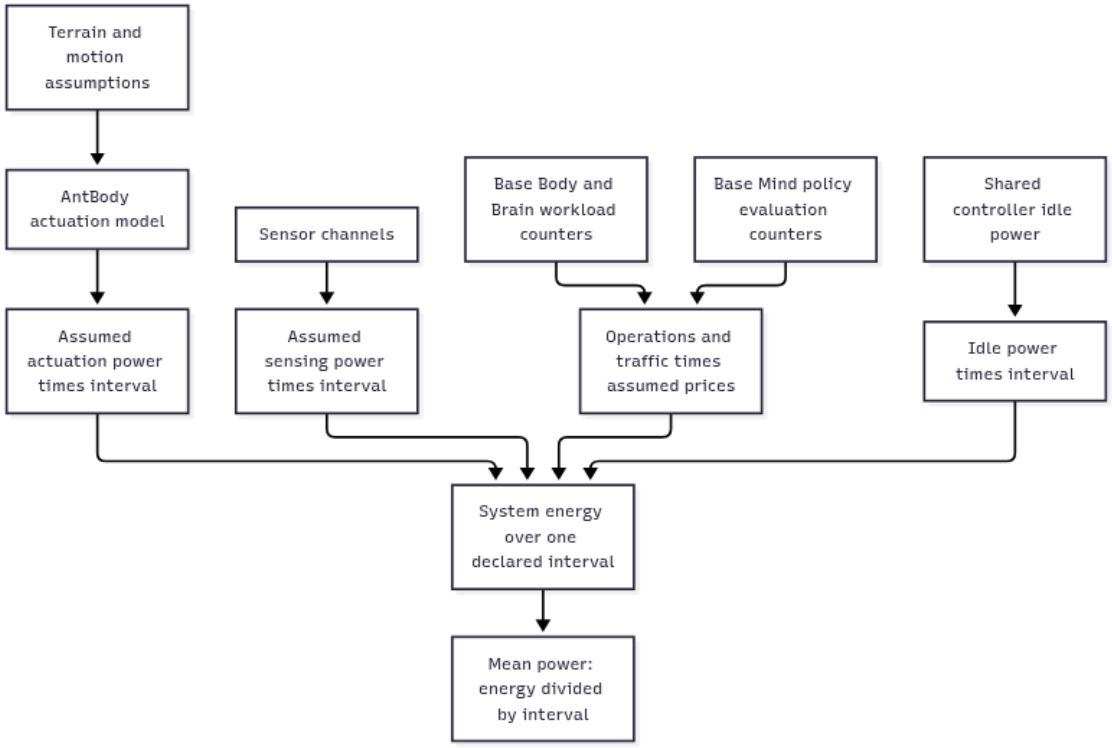


Figure 5: Common-Interval Analytical Energy Accounting: Modeled physical and compute terms share one interval and one idle contribution

Table 3: Coefficient Provenance

Input	Units	Provenance and interpretation
FLOP price	pJ/FLOP	Illustrative manifest assumption
SRAM/DRAM traffic prices	pJ/byte	Illustrative manifest assumptions; no device fit
Spike price	aJ/spike	Illustrative assumption; not whole-system measured energy
Idle power	W	Assumed load; once in a shared-controller sum
Actuator power	W/joint	Assumed electrical load, not dynamometer output
Sensor power	W/channel	Assumed load, not calibrated per-sensor telemetry

5.7 Transport and Physical References

Dimensionless cost of transport is $\mathrm{CoT} = E/(mgd)$ for energy E and displacement d covering the same traversal. The example's distance does not have a matched modeled locomotion interval, so **CoT is unavailable**, not a measured or inferred locomotion efficiency. At fixed distance, longer duration can add idle energy; equal-speed comparisons must scale distance with time.

Landauer's reference is $k_B T \ln 2$ per erased bit under the stated thermodynamic assumptions (Landauer 1961), not per FLOP or per decision. A matched erasure count and temperature are needed before comparing totals. Carnot's $1 - T_c/T_h$ applies to a heat engine between reservoirs; it is not a generic motor-efficiency ceiling. Shannon capacity requires a channel/noise model (Shannon 1948), not just a sensor-channel count. No optimization gap follows by dividing incomparable quantities.

5.8 Proposed Measurement and Calibration

Device counters and external meters should be time-aligned and accompanied by raw traces, calibration records, measurement boundaries, and held-out workloads. Report genuine independent replicates and parameter uncertainty separately from closed-form repetition and bootstrap Monte Carlo effort. These are proposed protocols, not completed measurements. See Methods and the external-data integration guide.

5.9 Units and Conversions

$1\,\mathrm{W} = 1\,\mathrm{J/s}$; $1\,\mathrm{mJ} = 10^{-3}\,\mathrm{J}$; $1\,\mathrm{pJ} = 10^{-12}\,\mathrm{J}$; $1\,\mathrm{aJ} = 10^{-18}\,\mathrm{J}$. A platform mass of $0.001\,\mathrm{kg}$ is $1\,\mathrm{g}$, whereas $1\,\mathrm{mg} = 10^{-6}\,\mathrm{kg}$.

6 Conditional Model Scaling and Limits of Interpretation

6.1 Descriptive Analytical Scaling

Parameter sweeps describe the selected closed-form models. They do not measure physiological scaling, solver convergence, or hardware latency. Current fit coefficients and intervals are reported only in the completed run's `Generated.md`; no fixed exponent or goodness-of-fit value is copied here.

A finite-range power-law fit $y = ax^b$ summarizes a curve over its stated range. A large b does not convert a power law into an exponential, and a nearly constant baseline-inclusive response does not imply zero computational or mechanical cost. Compute-only and baseline-inclusive curves answer different questions and must be labeled separately.

6.2 AntBody: Solver and Joint Dependence

In the selected base model, joint count affects both contact work and quadratic control work. The `pgs` iteration rule is $\min(50, \max(10, C))$; contact work is proportional to contact count times that iteration count at fixed J. Its piecewise operation counts are not a measured $C^{1.5}$ solver law. The other supported models are `si` and `ldlt`, not LCP/MLCP aliases. Their formula differences alone establish neither an accuracy ranking nor a preferred contact threshold.

Adding a fixed baseline can flatten energy variation relative to compute-only scaling. The manifest baseline is an assumed 0.50 W, not 50 mW. Morphological complexity is not physically free: mechanics, sensing, and hardware utilization must be considered on the same time basis. A nominal 100 Hz control rate is an input; real-time feasibility needs measured latency and deadline-miss distributions on identified hardware.

6.3 AntBrain: Activity and Connectivity Assumptions

At fixed N_{KC} and ρ, base channel processing grows with K and channel state traffic grows with KH. A fixed sparse-population term can dominate those increments over a finite range, but there is

no general parameter-independent neural energy law. The `random` and `structured` operation models use different assumed connectivity costs; neither is a measured ant topology. There is no supported biological-over-small-world efficiency comparison.

The example activity fraction $\rho = 0.02$ is an illustrative assumption. Lower modeled activity reduces some counters, but representational capacity, information loss, learning quality, and task performance are not evaluated by these counters. A sparsity optimum requires a task and data, not energy alone.

6.4 AntMind: Exact Trees Versus Bounded Policies

The selected base model sums tree nodes through horizon H_p with branching factor B. For $B > 1$, exact enumeration grows exponentially in horizon. Enhanced Mind is a distinct policy-count model with a uniform finite budget (default 1000 policies), or exact leaf enumeration when `max_policies=None`. A capped model can saturate its policy-count contribution; it does not exhibit the uncapped tree's growth by necessity. No universal 15-step real-time boundary follows from either model.

Inverse parameter values used in energy–proxy plots are declared mathematical proxies, not measured dexterity, accuracy, reward, or throughput. A Pareto frontier requires explicit objective directions and exclusion of dominated points; a sorted curve alone is not evidence of optimality.

6.5 Critical Behavior: Not Established

The released analytical workflow supplies no evidence of a second-order transition at 20 contacts, a first-order transition at activity fraction 0.02, or a planning divergence at horizon 15. Earlier numerical critical exponents (0.67, 0.8, 1.8), their intervals, a tenfold iteration jump, hysteresis, phase coexistence, universality classes, and finite-size critical-point shifts are not retained as results. The selected memoryless closed-form sweeps do not test path dependence or hysteresis.

A future critical-behavior study would need an explicit dynamical model and order parameter, bidirectional sweeps for path dependence, sufficient sizes and resolution, competing noncritical explanations, and independent observations. Piecewise counter rules, fit residuals, and policy-budget saturation should first be explained as model conventions, not physical phase transitions.

6.6 Design Interpretation

The framework can compare assumed operation/memory prices and workload dimensions while holding other inputs fixed. Such comparisons motivate device-specific profiling, sensor compression, activity-aware processing, and bounded planning experiments. Compute-aware stopping is supported as a distinct motion-planning research direction (Sudhakar et al. 2020), not demonstrated behavior of this analytical sweep. No hardware-independent savings, biological superiority, or universal scaling collapse is inferred.

7 Methods for Analytical Workload and Energy Estimation

7.1 Methodological Framework and Evidence Classes

This study uses deterministic analytical workloads, not a calibrated energy-measurement campaign. Retained quantities have the following provenance:

- **Assumptions:** workload dimensions, update rates, solver/pattern choices, memory splits, mechanical and sensor power coefficients, and digital energy prices in the manifest.
- **Analytical model outputs:** counters, coefficient-based energy estimates, descriptive fits, and generated plots from that manifest.
- **Literature context:** explicitly cited results from other systems; these do not calibrate this implementation.
- **Actual measured results:** none identified for the native example manifest, whose meter is null.
- **Proposed protocols:** the calibration, sensitivity, and external-validation procedures below; these have not been performed by the example workflow.

Software tests can establish implementation behavior against specified mathematical expectations. They do not establish biological realism, cross-platform timing, measurement accuracy, or publication readiness.

7.2 Computational Workload Modeling

7.2.1 Selected Base Workloads

The package orchestration module `antstack_core.orchestration.complexity_energetics` selects the base closed-form Body, Brain, and Mind functions. A window of duration t at update frequency f contains $n = \lfloor ft \rfloor$ modeled updates. Per-update counters are multiplied by n; byte counters represent modeled traffic over updates, not automatically peak resident memory.

AntBody: `calculate_contact_complexity` accepts only `pgs`, `si`, and `ldlt`. They are operation-count models, not solver accuracy or convergence benchmarks. For `pgs`, the iteration convention is $I = \min(50, \max(10, C))$ and contact FLOPs are $IC(6J + 12)$. For `si`, FLOPs are $20CJ + J^3/3$. For `ldlt`, FLOPs are $(J + 3C)^3/3$. Base Body adds $10S$ sensor FLOPs and $20J^2$ control FLOPs per update. No special transition at 20 contacts is inferred from these formulas.

AntBrain: the supported patterns are `random` and `structured`. For $N_a = \lfloor \rho N_{KC} \rfloor$, their neural FLOP models are N_a^2 and $N_a \log N_{KC}$ respectively. Base Brain adds $100K$ channel FLOPs and $4KH$ channel bytes per update. These labels do not implement small-world, scale-free, or species-specific biological topologies. The configured $\rho = 0.02$ is an assumed activity fraction, not an inferred optimum.

AntMind: base planning uses canonical parameters B and H_p, with the exact node count

$$N_{\text{tree}} = \sum_{h=0}^{H_p} B^h. \tag{12}$$

Belief, policy, and evidence-bound counters scale with this node count and the configured state/action dimensions. H_p is measured in decision steps, not seconds. The model counts work; it does not execute or assess policy decisions.

7.2.2 Separate Enhanced Models

Enhanced Body adds a more detailed sensor/control decomposition; enhanced Brain separately represents AL, MB, CX, and plasticity counters. These must not be used to interpret a base-model sweep without identifying the change of model. Enhanced Mind uses $P = \min(B^{H_p}, M)$ policies for a finite `max_policies=M` (default 1000), uniformly across horizons; `max_policies=None` uses $P = B^{H_p}$.

This leaf-policy convention differs from the base model's sum of tree nodes. Bounding policy counts does not demonstrate preserved decision quality. Hierarchical overhead is a model assumption, not an observed acceleration.

7.3 Energy Estimation and Time Accounting

The manifest is the numerical input authority for illustrative coefficients. Compute energy over a window is

$$E_{\text{compute}} = 10^{-12}(N_F e_F + V_S e_S + V_D e_D) + 10^{-18} N_{\text{spk}} e_{\text{spk}}, \tag{13}$$

where FLOP/byte prices are in pJ and the spike price is in aJ. Baseline energy is $P_0 t$. Mechanical and sensing power estimates must be integrated over the same window before any sum or cost-of-transport calculation. A component must appear only once within the declared system boundary; separate module estimates with their own baseline are not automatically a shared-controller total.

Each result row reports module-isolated compute plus its own idle contribution over that workload window. These rows are not additive shared-controller system energy. The separate per-decision system table uses the same selected base compute models, assumed physical actuation/sensing, and one shared idle baseline at the configured update rate. Cost of transport is unavailable because the supplied distance has no matched modeled locomotion interval.

The generated report must identify whether a statistic is a workload-window energy, a total over repeats, a mean over repeats, or energy per decision. For $n > 0$, window energy divided by n yields energy per modeled update; power is window energy divided by t. A 0.5 s workload at 100 Hz contains 50 updates, not one decision. At 0.50 W, baseline is 0.25 J over that window and 0.005 J over 10 ms. Do not compare these as equal intervals.

7.4 Descriptive Scaling Fits and Resampling Limits

Log–log regression fits a descriptive finite-range power law $y = ax^b$ to positive finite paired values. A large exponent b is not itself an exponential law $y = a\exp(cx)$, and a goodness-of-fit score is not physical validation. Report the parameter range, response boundary (compute-only versus baseline-inclusive), model family, and fit limitations with each result. For a constant response, the descriptive exponent is zero but the variance-based goodness-of-fit R^2 is undefined (reported as `None`/unavailable), not evidence of a perfect fit or superiority.

The example requests 100 closed-form repeats and 1000 bootstrap resamples at confidence level 0.95. The former repeat identical formulas for fixed inputs; the latter resample the supplied values. Neither count is an independent empirical sample size, and neither guarantees power greater than 0.8. Identical values have a degenerate resampling interval; no artificial width should be added. Parameter uncertainty and device error require separate justified input distributions or measurements.

Sobol indices, Morris screening, Bayesian posterior calibration, held-out hardware prediction error, and cross-laboratory replication have not been established by this manifest. A future sensitivity analysis must specify parameter distributions, dependence assumptions, sampling design, and evaluated responses. A future power analysis must specify an estimand, independent sampling unit, effect size, variance, test, and error rates.

7.5 Proposed Measurement and Calibration Protocol

Before describing coefficients as calibrated, identify a device and measured rails, run operation/memory/spike microbenchmarks with documented instrumentation, record temperature and

utilization, and fit coefficients with uncertainty. Validate on independent held-out workloads. Avoid transferring processor-generation trends (Koomey et al. 2011) into a device-specific calibration claim.

RAPL/NVML telemetry and external meters are possible instruments, not evidence that a measurement was taken. Record counter range/wrap behavior, monotonic timestamps, sampling bandwidth, synchronization error, firmware, and meter calibration records. Specify whether sensing, actuation, idle power, and host computation are included. No WT310 accuracy, 100 Hz meter synchronization, dynamometer result, 7 nm circuit calibration, or attojoule-scale measurement resolution is claimed here.

Independent ant validation additionally needs species/caste, colony and individual identity, mass definition, temperature, behavioral context, raw records, licenses, and held-out comparisons. General robot modeling literature (Jaramillo-Morales et al. 2020) and compute-aware planning (Sudhakar et al. 2020) motivate the approach but do not validate these ant or hardware models.

7.6 Reproducibility and Publication Boundary

Retain the exact manifest, package/environment versions, invocation, model selection, result identity, and artifact provenance. `Generated.md` is the current numerical-result section only when produced by the same completed governed run as the selected figures and tables. Retained older PDFs, concatenations, and plots are historical artifacts, not proof of a fresh successful run. Rendering, cross-reference checks, and numerical tests are separate acceptance steps; none substitutes for empirical calibration.

8 Analytical Complexity and Energy Results

These are analytical/synthetic workloads, not independent biological or hardware observations.

8.1 Per-Workload Estimated Energy

Each row includes that module's compute and isolated idle over its workload duration. These isolated rows must not be summed as a system total; locomotion is excluded.

Workload	Mean (J/window)	95% CI Low	95% CI High	N
body	0.25000291	0.25000291	0.25000291	100
brain	0.25004641	0.25004641	0.25004641	100
mind	271.14218	271.14218	271.14218	100

Component	Energy (mJ / 10 ms)
body	5.817664e-05
brain	0.0009281049
mind	5417.8436
Body actuation	360
Body sensing	12.8
Shared idle	5
Modeled system total	5795.6445

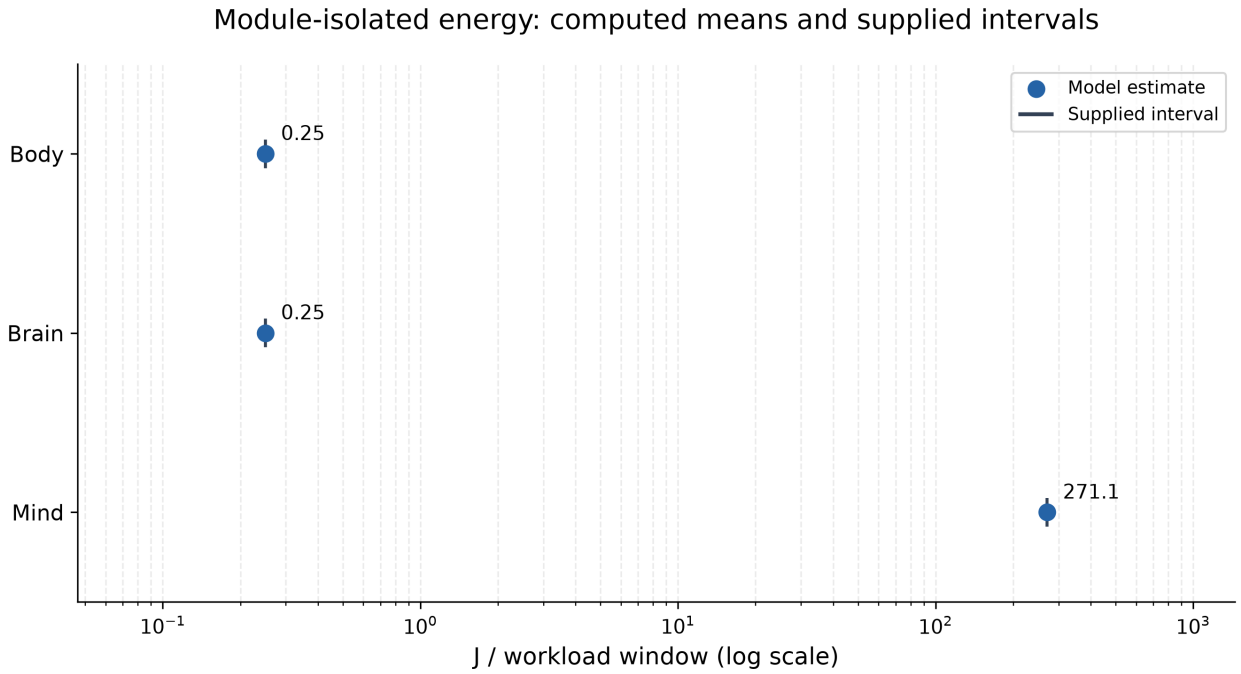


Figure 6: Module-Isolated Energy: Mean module-isolated compute plus idle, with 95% bootstrap intervals for the mean. Repeating deterministic values does not establish empirical precision; positive wide-range values use a labelled logarithmic axis.

Table 4: Common-Window Complexity

Workload	FLOPs / 10 ms	SRAM bytes / 10 ms	DRAM bytes / 10 ms	Spikes / 10 ms
body	26320	6246.4	1561.6	0
brain	23619.778	389491.2	43276.8	100
mind	3.8883771e+12	1.7637999e+11	7.5591425e+10	0

8.2 Cost of Transport

Unavailable: compute-window energy is not locomotion energy, and no matching modeled locomotion interval/distance is supplied by this producer.

8.3 Raw Results

Results CSV

9 Analytical Results and Evidence Boundaries

9.1 Provenance and Conditional Status of Generated Results

The numerical result section is `Generated.md`, produced from the native manifest by the package CE analysis. Its tables and figures must belong to one completed governed run. This prose intentionally

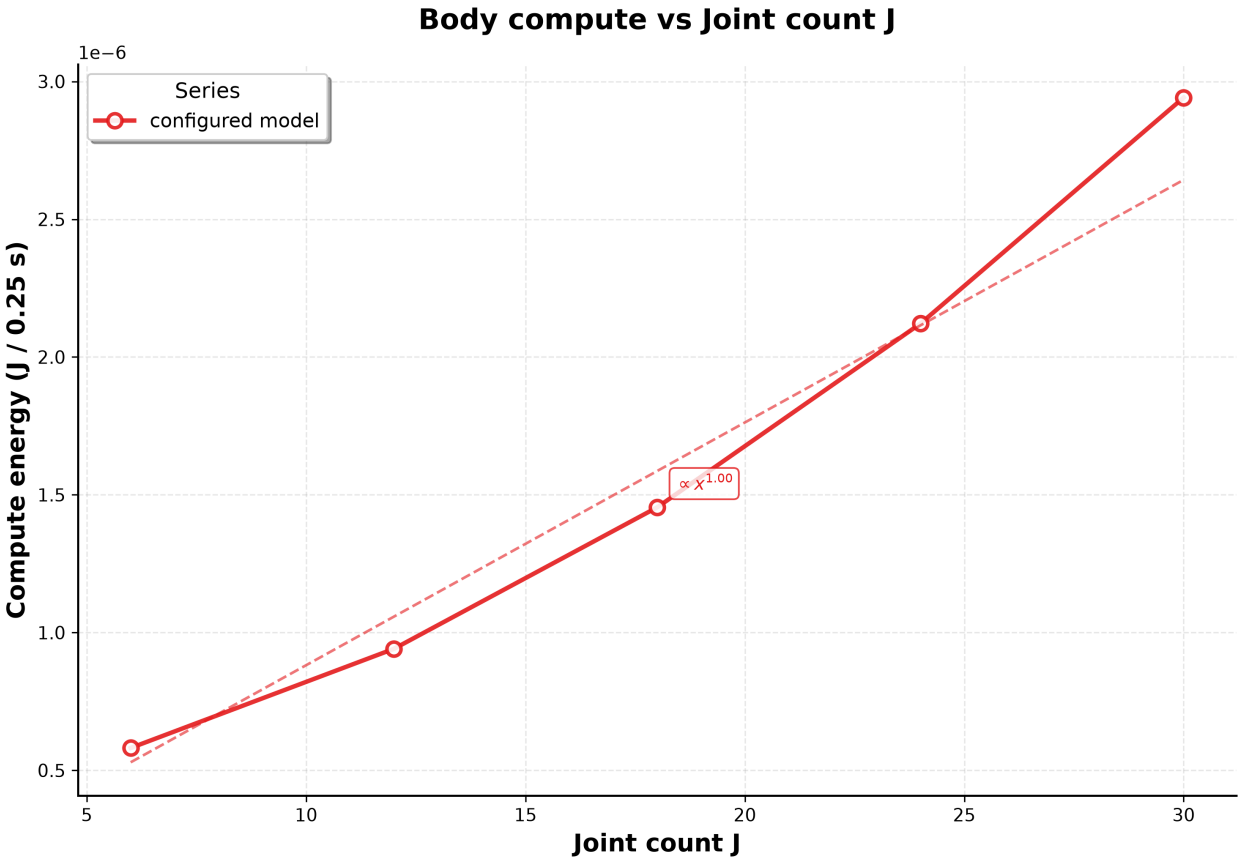


Figure 7: Body configured compute curves: Calculated values over a 0.25 s compute-only window; lines connect model outputs, not empirical replicates. No fixed biological scaling law is inferred.

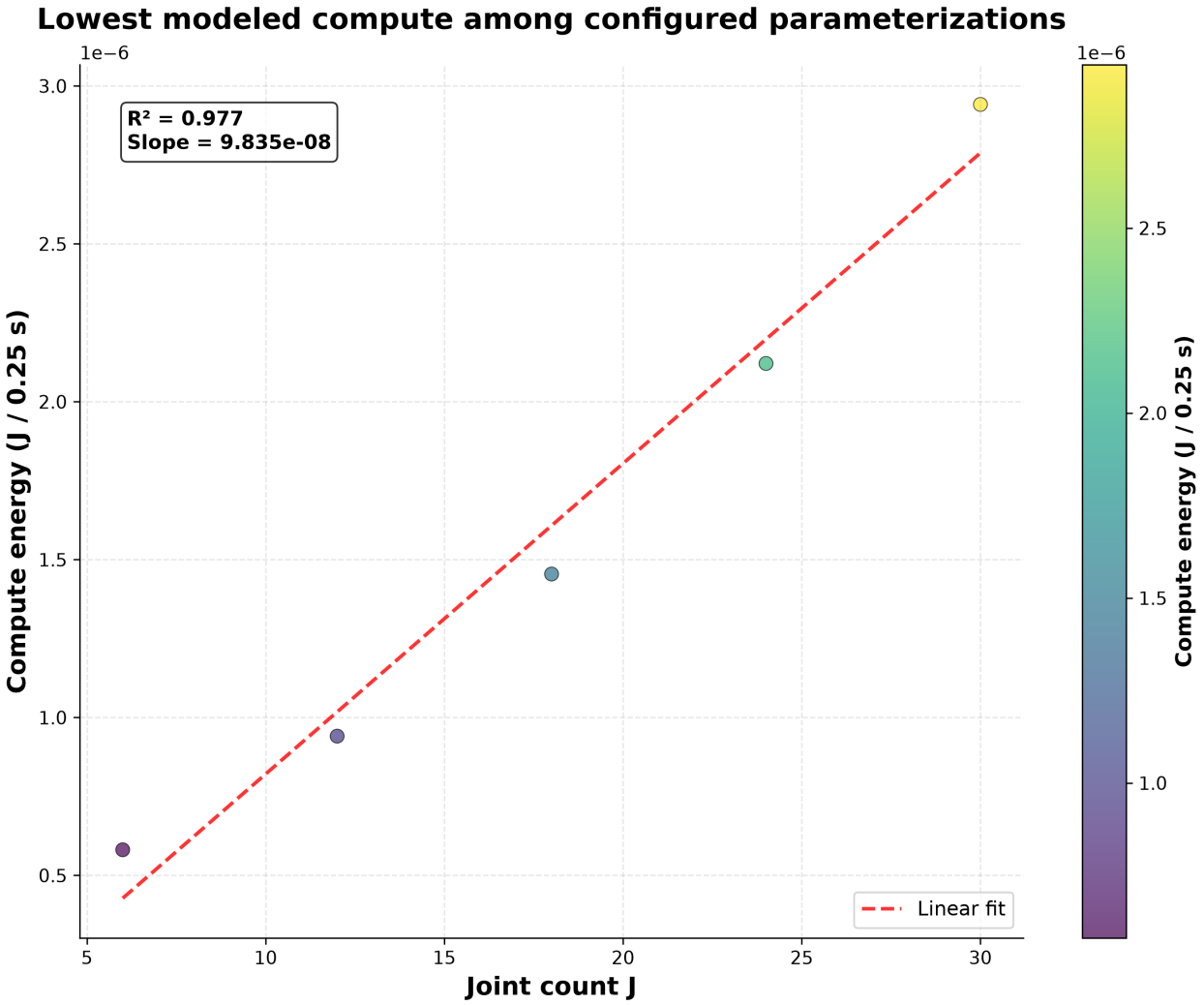


Figure 8: Body lowest configured compute points: Calculated values over a 0.25 s compute-only window; lines connect model outputs, not empirical replicates. No fixed biological scaling law is inferred.

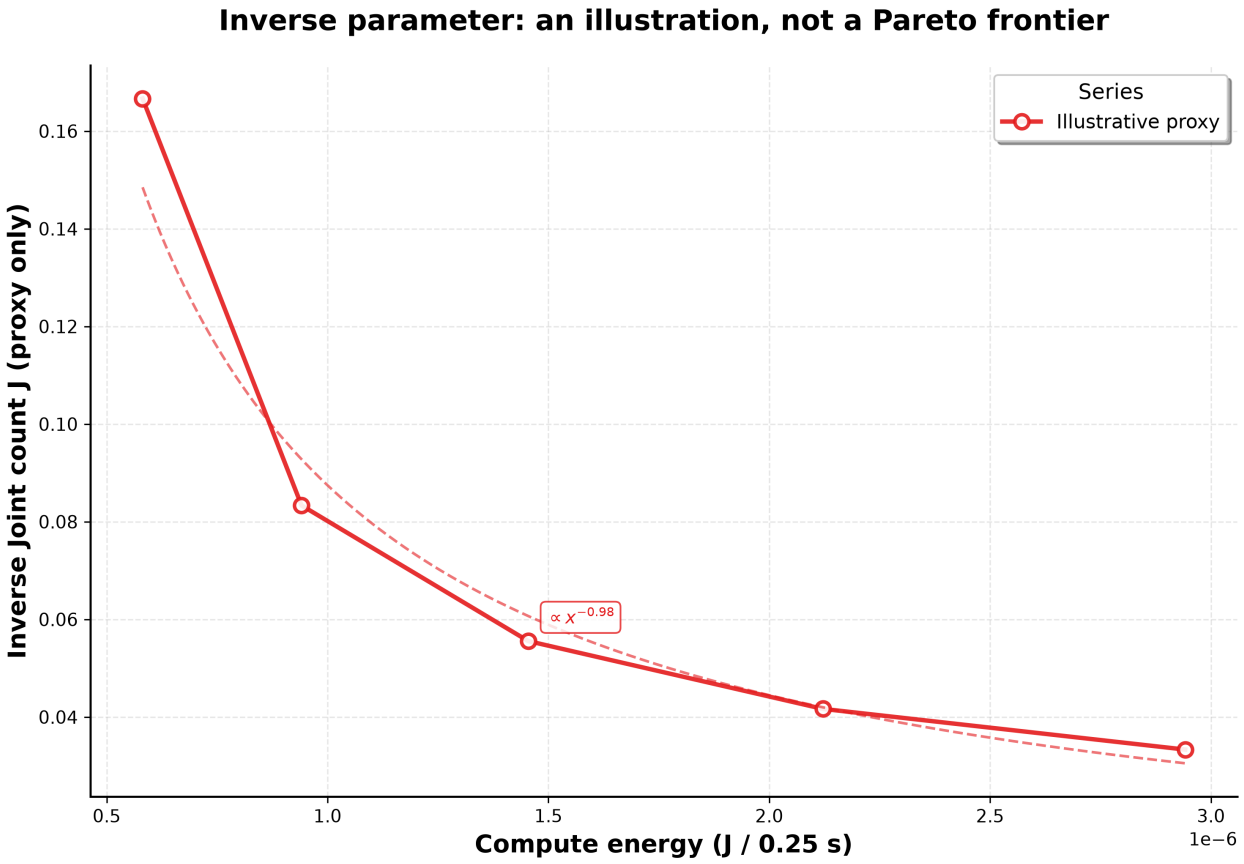


Figure 9: Body inverse-parameter illustration: Calculated values over a 0.25 s compute-only window; lines connect model outputs, not empirical replicates. The inverse-parameter curve is a proxy, not measured performance or a nondominated frontier.

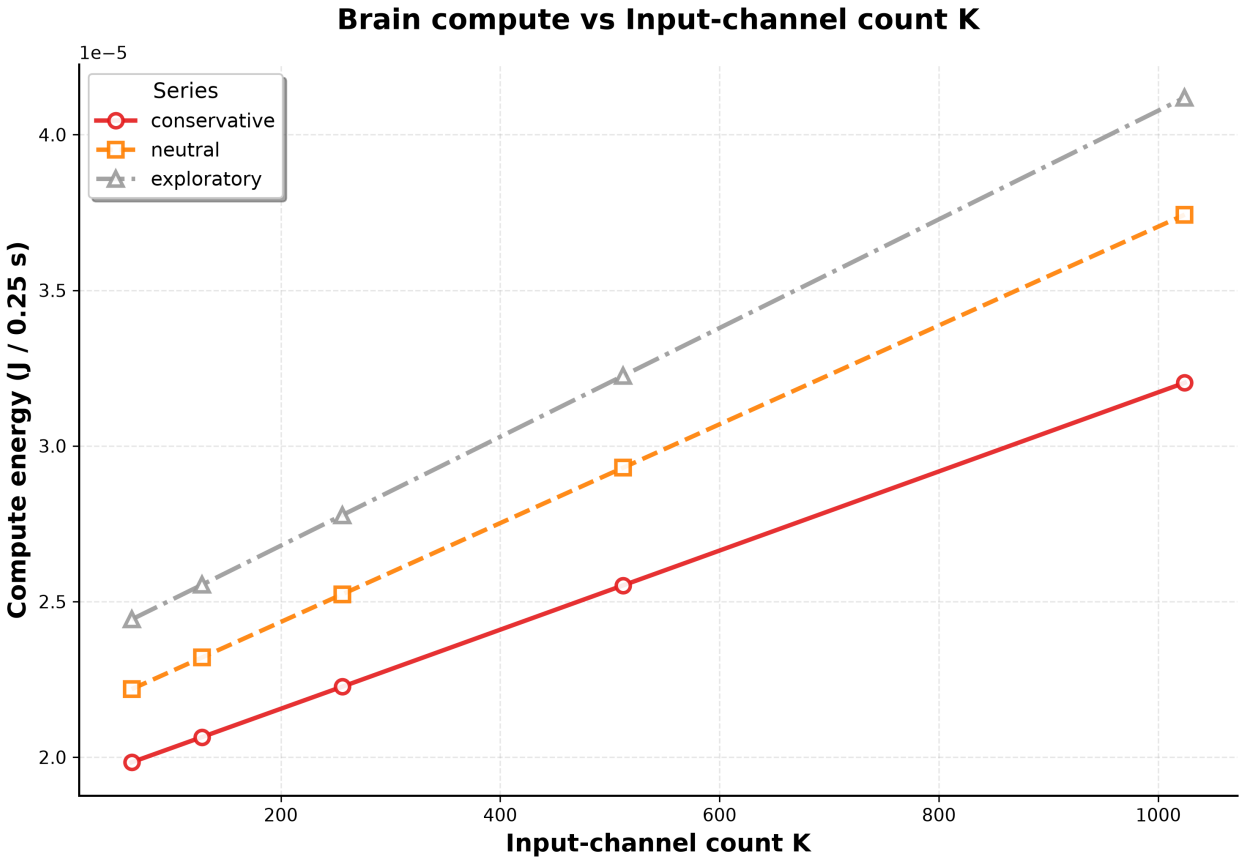


Figure 10: Brain configured compute curves: Calculated values over a 0.25 s compute-only window; lines connect model outputs, not empirical replicates. No fixed biological scaling law is inferred.

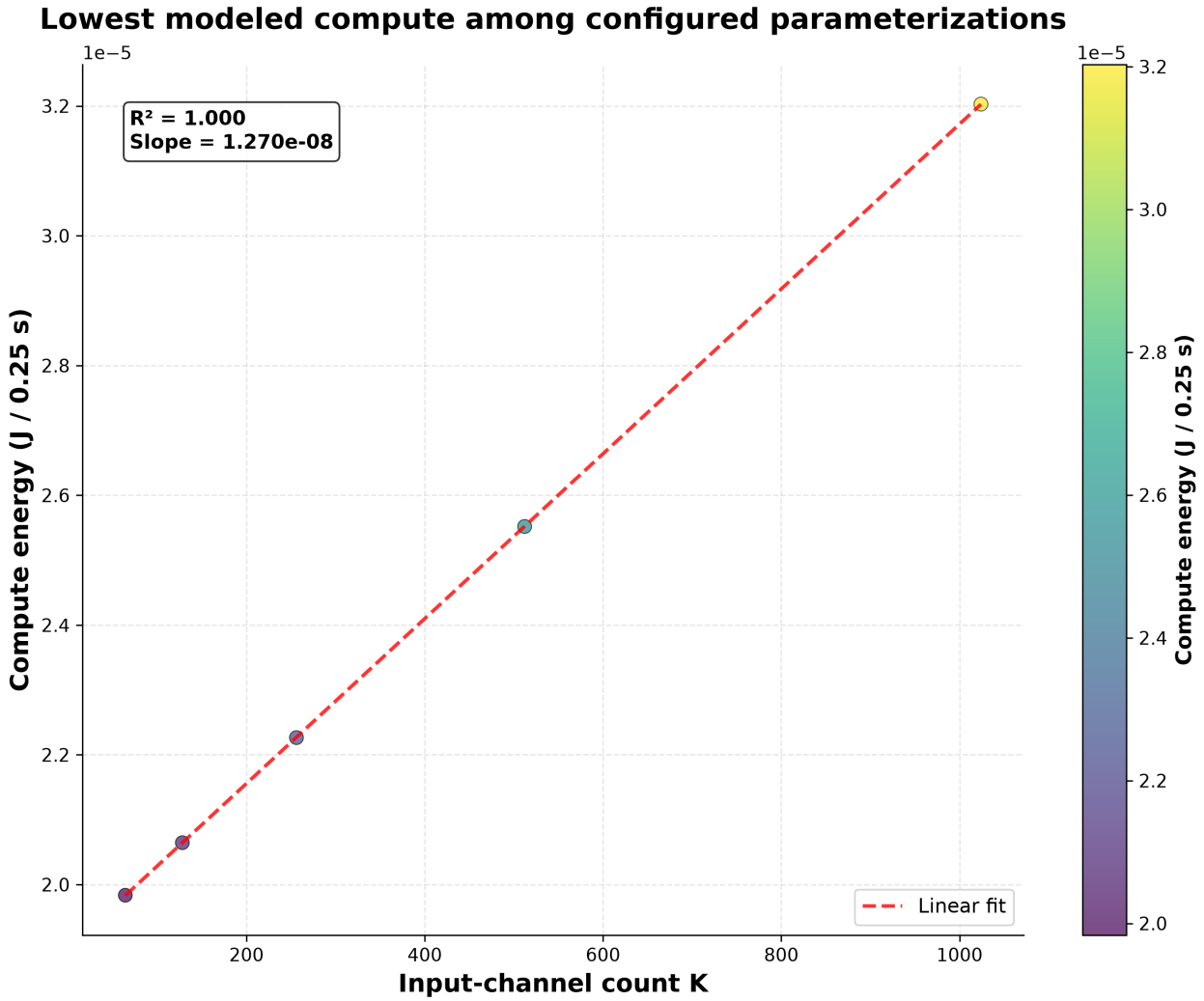


Figure 11: Brain lowest configured compute points: Calculated values over a 0.25 s compute-only window; lines connect model outputs, not empirical replicates. No fixed biological scaling law is inferred.

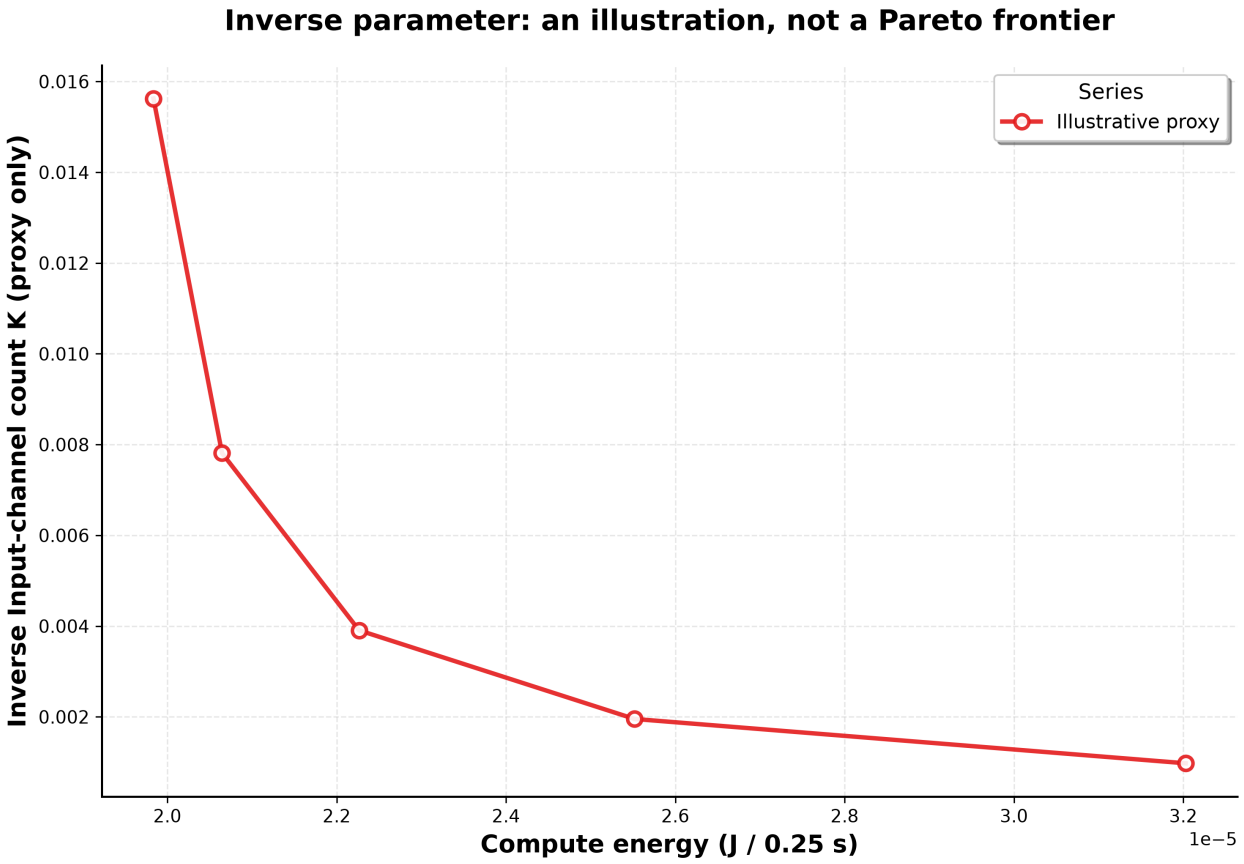


Figure 12: Brain inverse-parameter illustration: Calculated values over a 0.25 s compute-only window; lines connect model outputs, not empirical replicates. The inverse-parameter curve is a proxy, not measured performance or a nondominated frontier.

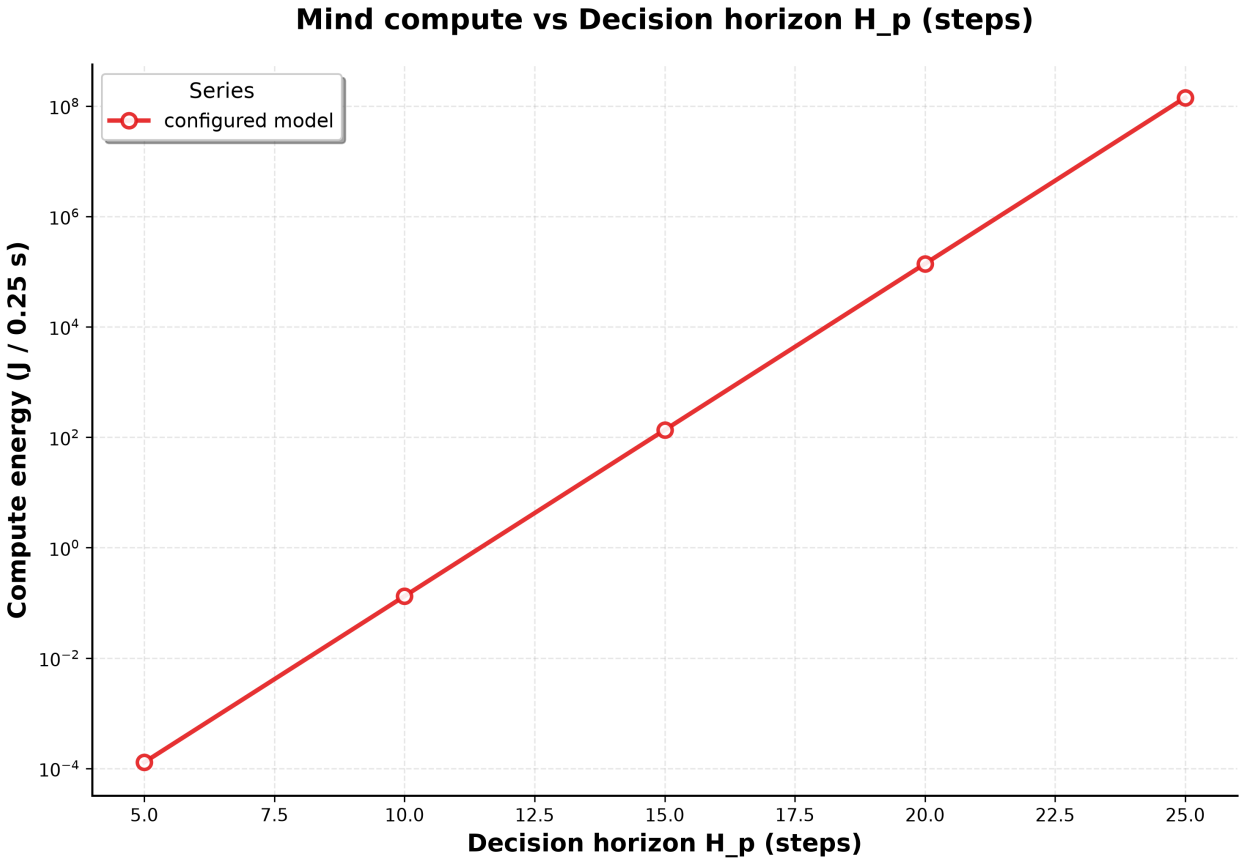


Figure 13: Mind configured compute curves: Calculated values over a 0.25 s compute-only window; lines connect model outputs, not empirical replicates. No fixed biological scaling law is inferred.

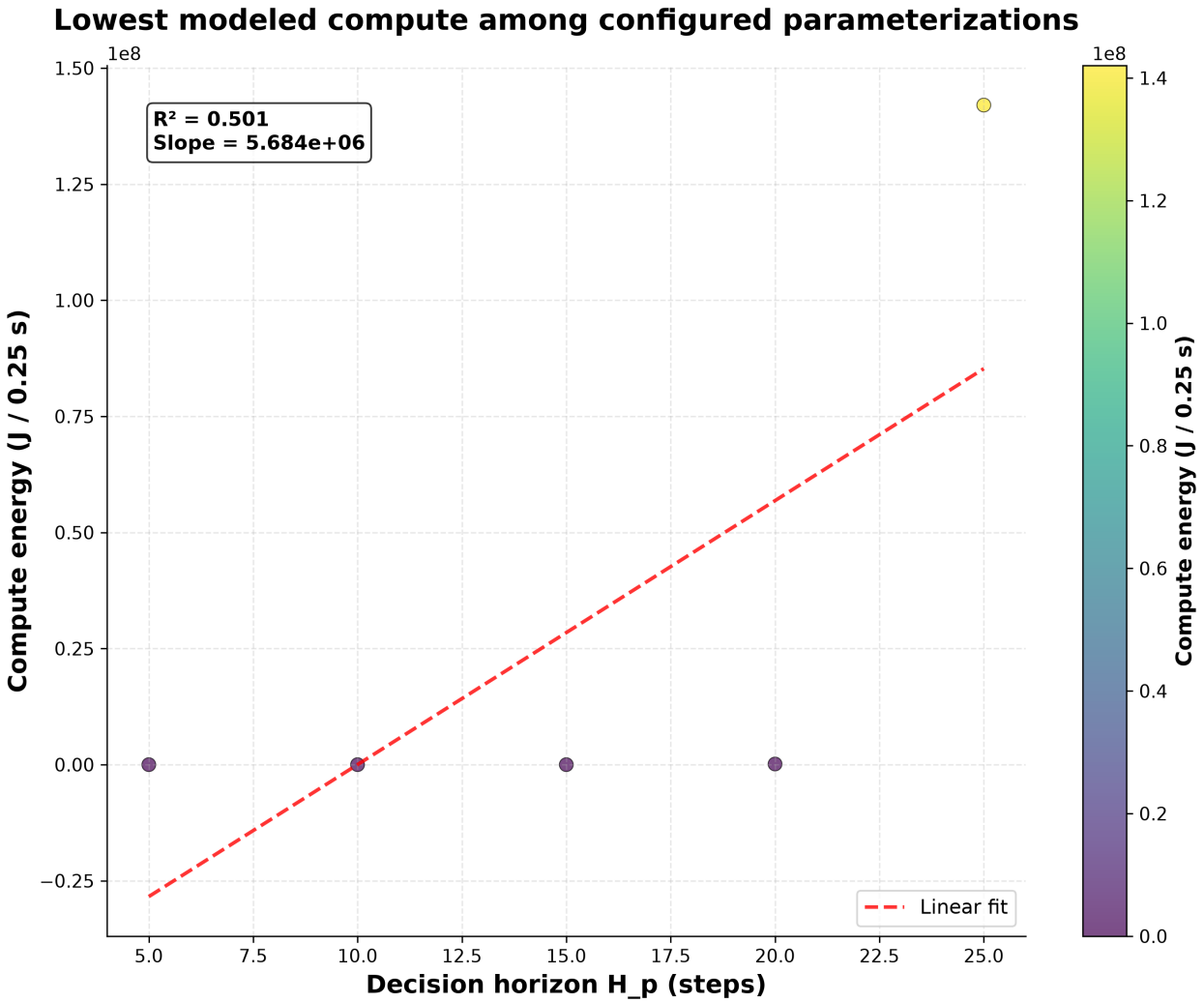


Figure 14: Mind lowest configured compute points: Calculated values over a 0.25 s compute-only window; lines connect model outputs, not empirical replicates. No fixed biological scaling law is inferred.

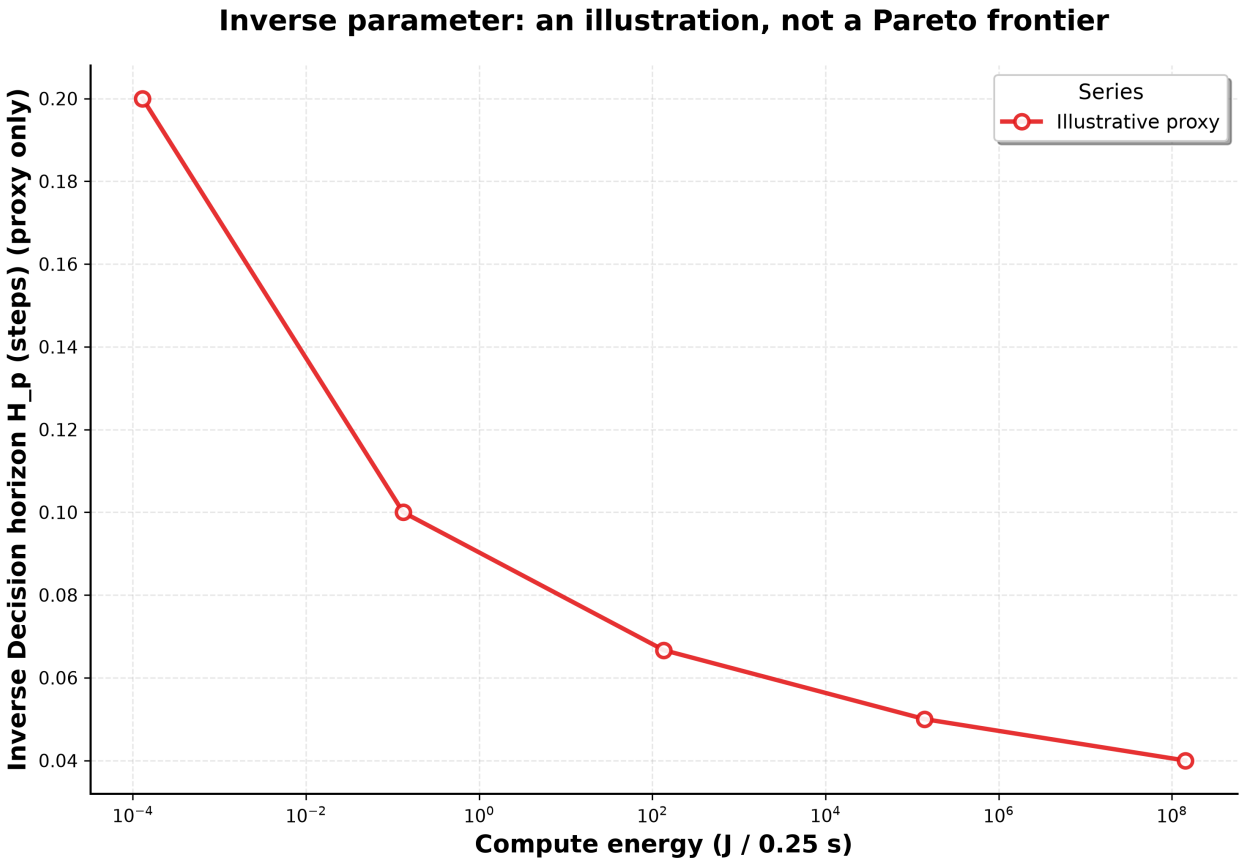


Figure 15: Mind inverse-parameter illustration: Calculated values over a 0.25 s compute-only window; lines connect model outputs, not empirical replicates. The inverse-parameter curve is a proxy, not measured performance or a nondominated frontier.

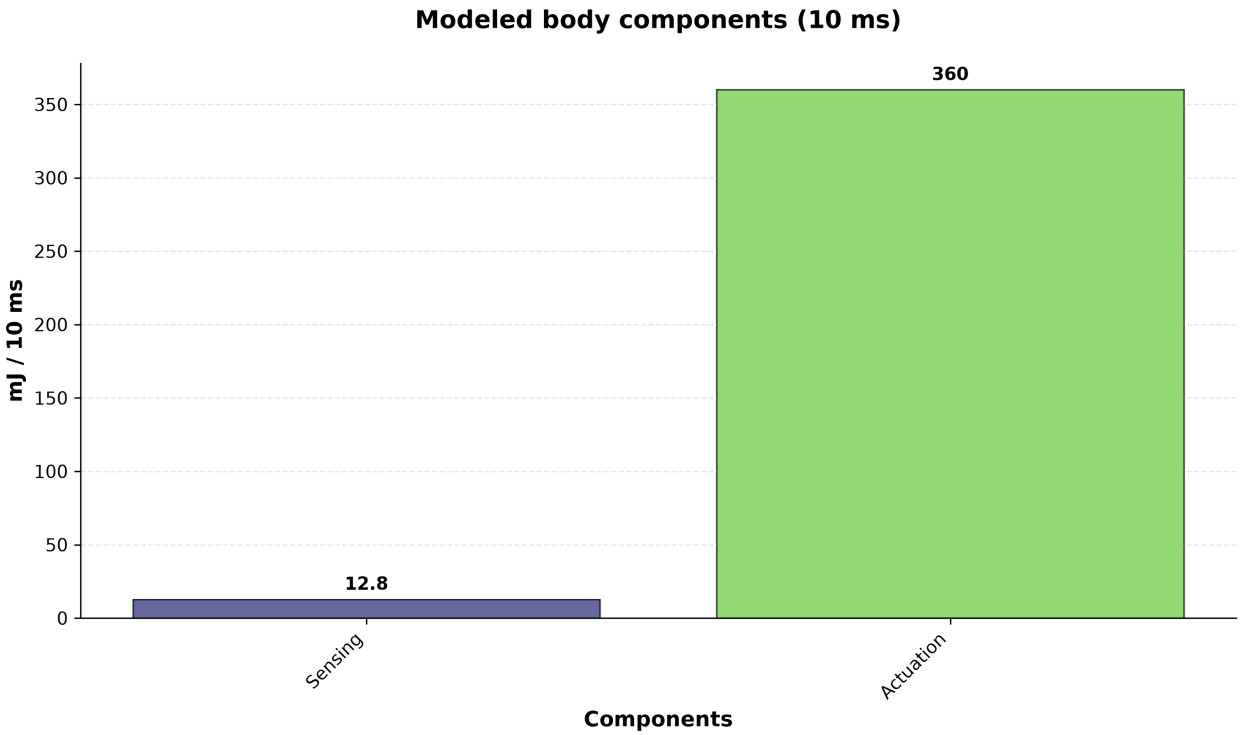


Figure 16: Body Energy Partition: Separate sensing and actuation assumptions integrated over the common 10 ms interval, not the compute-only workload row.

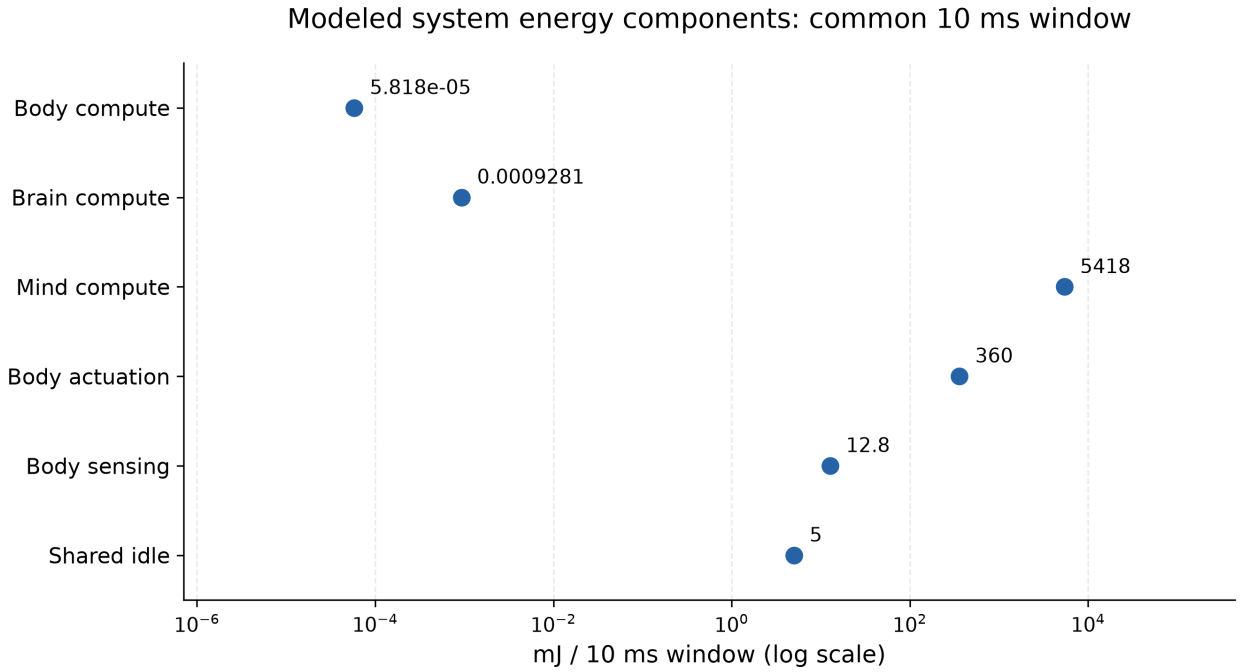


Figure 17: Common-Window Compute Accounting: Selected workload counters over a common 10 ms window, inverse maximum configured update rate, with shared idle counted once. Integer update flooring can omit slower modules; this is not their time-averaged load or necessarily one decision each. Sensing and actuation are separate assumptions, not observed locomotion.

does not duplicate fitted exponents, confidence intervals, energies, component percentages, or cost-of-transport values that would become stale when the manifest or implementation changes. Checked-in generated material and retained PDFs are historical until the release owner generates and inspects a coherent fresh result set.

The example manifest uses deterministic closed-form workloads and a null meter. Thus its reported energy is an **analytical model output conditional on assumed coefficients**, not an actual measured result. Repeating those formulas 100 times and using 1000 bootstrap resamples does not create independent physical observations, calibrate coefficients, or demonstrate statistical significance.

9.2 Reading the Generated Outputs

- **Body:** identify `pgs`, `si`, or `ldlt` and the base-model dispatch. A joint sweep includes contact and quadratic control terms; no fixed near-zero exponent or 20-contact timing boundary is asserted.
- **Brain:** identify the supported `random` or `structured` pattern, activity fraction, population size, and channel range. The output is not a comparison of biological, small-world, or scale-free networks.
- **Mind:** the base model uses the exact tree-node sum with B and H_p; the separate enhanced model's finite policy budget must not be attributed to a base sweep. A finite power-law approximation to tree growth is descriptive, not proof of a critical exponent or a 15-step feasibility limit.
- **Uncertainty:** say what was resampled and which statistic the interval bounds. A total-over-repeats bar needs an interval for the total, not the mean. Zero-width intervals for identical deterministic inputs are legitimate and do not describe device uncertainty.
- **Energy–proxy comparisons:** identify objective directions and the proxy definition; neither an inverse dimension nor a sorted curve demonstrates measured task performance or a nondominated optimum.

9.3 Unit and Accounting Interpretation

Window energy, energy per decision, and mean power are different quantities. For duration t and frequency f, normalize by the actual modeled decision count $\lfloor ft \rfloor$ when nonzero. Match the integration window for compute, sensing, mechanics, and baseline before adding them. Rows are module-isolated compute-plus-idle estimates and must not be summed as a system total. The per-decision system table derives the same base compute models, adds assumed physical actuation and sensing, and allocates one shared idle baseline.

The example's 0.50 W baseline contributes 0.005 J (5 mJ) in a 10 ms interval. This is a dimensional illustration of an **assumed input**, not a generated or measured module result. Body compute must not disappear from a body breakdown, and a compute-only component sum must not be renamed actuation energy. A conceptual Mind layer is not physically energy-free: its implemented policy-evaluation counters incur modeled compute energy wherever they run.

Cost of transport is $E/(mgd)$, with all energy in the numerator integrated over the same traversal as distance d. Declare whether E includes actuation, sensing, compute, and idle load. The native run reports CoT as unavailable: its supplied distance has no matched modeled locomotion interval. A workload-window compute estimate alone is not whole-body locomotion CoT. At fixed speed, changing duration also changes distance; at fixed distance, a longer duration can legitimately change

CoT because idle energy accumulates. A mass of 0.001 kg equals **1 g**, not 1 mg; 1 mg is 10^{-6} kg. No species identity follows from a nominal platform mass.

9.4 Physical References Are Not Efficiency Validation

Landauer's $k_B T \ln 2$ is a reference per logically irreversible erased bit under its thermodynamic assumptions (Landauer 1961). At approximately room temperature it is of order 10^{-21} J/bit. It is not a calibrated per-FLOP energy and cannot be divided directly into total workload energy without an explicit compatible erasure count and boundary. Separate mechanical work and bit-erasure references are not interchangeable minima.

A value below a stated lower reference is an accounting/applicability question, not evidence of near-optimality. The manuscript makes no retained 4.2×10^8 or 2.7×10^6 optimization-factor claim, no below-bound efficiency conclusion, and no universal actuation-dominance percentage.

9.5 Biological Comparison: Data Required

No independent raw ant energetics or neural-energy dataset has been identified for this release's native workflow. The earlier *Formica rufa* metabolic rates, ant CoT range, mushroom-body power range, universal mass/neuron exponents, muscle/motor comparisons, and derived biological-to-robotic improvement ratios are not retained as established results. The named literature and configured robot assumptions did not supply traceable matched evidence for those claims.

A biological comparison requires a primary dataset with species/caste, wet/dry mass, temperature, locomotor speed and distance, metabolic/measurement boundary, sample independence, and uncertainty. Robot power-model studies (Jaramillo-Morales et al. 2020) are relevant methodological context, not ant validation. Dataset ingestion and a held-out comparison are prerequisites, as described in the external-data guide; passing source or render checks does not satisfy them.

9.6 Proposed Follow-On Measurements

Measure workload latency and deadline misses on a named device before claiming 100 Hz feasibility. Calibrate operation/memory prices and verify complete system accounting on held-out workloads before claiming prediction accuracy. Test sparse representations and policy budgets against task outcomes before claiming an accuracy–energy advantage. Sobol/Morris analyses, power calculations, and critical-behavior experiments remain proposed studies rather than completed results of this run.

10 Model Interpretation, Accounting Limits, and Proposed Validation

10.1 Interpretation of the Analytical Framework

Body, Brain, and Mind emphasize different configured dimensions: contact/control operations, sparse-population/channel operations, and policy-tree or policy-budget work. This distinction is useful for designing experiments, but it is not evidence that all embodied systems occupy three universal energy regimes. Numerical conclusions are conditional on the selected base model, manifest assumptions, and complete accounting boundary shown in `Generated.md`.

10.1.1 Information-Theoretic References

Shannon capacity (Shannon 1948) needs bandwidth and a noise model; the present counters do not determine achieved bits per second. The earlier claim that AntBrain reaches 0.1% of capacity is not supported. Landauer's erased-bit reference (Landauer 1961) requires a compatible erasure count and temperature. The former 4.2×10^8 neural optimization factor is not a retained efficiency result or hardware target.

10.1.2 Biological Inspiration Versus Comparison

Insect-inspired architecture motivates sparse representations and local processing. It does not demonstrate that `structured` is a biological network or that it outperforms small-world/scale-free alternatives. The implementation supports only `random` and `structured` neural counter models. The claimed measured 15–25% biological advantage, 16-fold sensory expansion at constant energy, and 60–80% event-driven savings are not retained as established outcomes. Task quality and independent energy evidence would be needed for those comparisons.

No matched biological CoT, metabolic-rate, muscle/motor, or mushroom-body-power dataset has been identified for this workflow. Therefore no biological-to-robotic improvement ratio or species-specific efficiency ceiling is inferred. Nominal platform mass and assumed joint power are engineering choices, not ant physiological measurements.

10.2 Module-Specific Design Questions

Body: solver choice changes assumed operations and memory traffic. The supported `pgs`, `si`, and `ldlt` formulas do not rank numerical accuracy, guarantee a 10 ms deadline, or justify switching at 20 contacts. Fixed baseline can dominate a compute-inclusive curve without making morphology mechanically free. Terrain and duty-cycle savings require a specified physical model or data rather than fixed percentages.

Brain: population activity, channel traffic, and memory prices offer dimensions for sensitivity studies. The example $\rho = 0.02$ is not a validated optimum. Reduced counters alone say nothing about preserved representation, learning quality, or task accuracy. Device-specific memory locality and event-driven execution should be evaluated on actual workloads.

Mind: the base exact tree grows with B and H_p, while enhanced bounded policies are a distinct model with an explicit uniform budget. Sampling and hierarchical organization motivate experiments but are not demonstrated quality-preserving accelerations. No universal horizon-15 barrier is established, and conceptual symbolic processing does not eliminate physical compute.

10.3 Accounting and Uncertainty Limits

Rows are module-isolated compute-plus-idle estimates over workload windows. The system per-decision table uses the same selected base compute models, assumed actuation/sensing, and one shared baseline. Different intervals and repeated idle allocation must not be confused with physical efficiency differences. CoT is unavailable because the supplied distance has no matched modeled locomotion interval.

Deterministic repeats and bootstrap resamples characterize supplied values, not independent device or animal variability. A degenerate interval is truthful for repeated constants. High fit scores and finite-range exponents are descriptive; they establish neither statistical power nor physical

critical behavior. Earlier contact/sparsity/horizon critical exponents and hysteresis claims lack corresponding dynamical evidence and have been removed as findings.

10.4 Research Directions and Proposed Protocols

1. **Device calibration:** measure operation/memory costs and idle rails, propagate parameter uncertainty, and validate on held-out workloads.
2. **Timing and task quality:** record latency/deadline misses and task outcomes under supported solver, sparsity, and policy-budget choices.
3. **Energy-aware planning:** compare an explicitly implemented compute-aware stopping rule against task-matched baselines, following the motivation of CEIMP (Sudhakar et al. 2020).
4. **Biological integration:** acquire licensed species/caste-specific raw records and use held-out colony/individual splits; do not transplant unrelated allometric laws or neural-power estimates.
5. **Multi-agent behavior:** specify field dynamics, independent replicates, and task metrics before fitting colony-size scaling. No coordination exponent or critical colony size is established by the current module sweeps.

A future Sobol/Morris study needs declared parameter distributions and a sampling design; a power analysis needs independent units, effect/variance assumptions, and a test. These are proposed analyses, not completed validations. Integration with external ecological resources is a research interface, not evidence of an already calibrated end-to-end simulator.

10.5 Conclusion

The contribution is a transparent analytical accounting workflow with explicit model choices and evidence boundaries. It can support implementation checks and conditional design comparisons. It does not establish biological superiority, hardware prediction accuracy, universal real-time limits, critical phenomena, independent replication, or publication readiness. Those conclusions require the measurements and task-specific evidence named above.

11 Acknowledgements

We extend our sincere gratitude to Marek P. Bargiel for his thorough and constructive peer review of this manuscript.

12 Resources and Checked Literature

12.1 Energy-Aware Robotics

Jaramillo-Morales, Dogru, Gomez-Mendoza, and Marques (2020), *Energy estimation for differential drive mobile robots on straight and rotational trajectories* (Jaramillo-Morales et al. 2020), describes a platform-specific model and validation. It is methodological context, not calibration of the Ant Stack.

Sudhakar, Karaman, and Sze (2020), *Balancing Actuation and Computing Energy in Motion Planning* (Sudhakar et al. 2020), introduces compute-energy-aware stopping in motion planning. The authors' primary publication page supplies the identity and scope; no unrelated conference DOI is attached.

12.2 Computing and Physical References

Koomey, Berard, Sanchez, and Wong (2011), *Implications of Historical Trends in the Electrical Efficiency of Computing* (Koomey et al. 2011), concerns historical trends, not device-specific operation prices for this manifest.

Landauer (1961), *Irreversibility and Heat Generation in the Computing Process* (Landauer 1961), supplies erased-bit thermodynamic context. Shannon (1948), *A Mathematical Theory of Communication* (Shannon 1948), supplies information-theoretic context. Neither reference validates a per-FLOP coefficient, whole-robot efficiency ratio, or achieved neural channel capacity.

12.3 Active-Inference Context

Friston, Rigoli, Ognibene, Mathys, Fitzgerald, and Pezzulo (2015), *Active inference and epistemic value* (Friston et al. 2015), supplies decision-theoretic context. Its variational/expected free energy is not automatically a physical energy quantity.

12.4 Bibliographic Provenance

Retained bibliography identities were checked against primary publisher-deposited DOI metadata or the authors' institutional publication record. The earlier duplicate/conflicting records were removed rather than assigned guessed identities. In particular, DOI `10.1038/s41467-020-16108-9` identifies Joshi and colleagues' *Accurate deep neural network inference using computational phase-change memory* (2020), not either former sparse-network/neuromorphic title. That unrelated paper is not used to support this manuscript. The former CEIMP DOI `10.1109/ICRA40945.2020.9196880` identifies a different modular-robot paper and has likewise been removed from the CEIMP citation.

Search topics such as hardware energy telemetry, locomotion profiling, sparse neural processing, sensitivity analysis, and animal tracking are research directions, not completed validation or evidence for quantitative claims. External datasets require raw-data provenance, units, species context, and a declared independent validation design before use as manuscript evidence.

13 Appendices

13.1 A. Energy Input Assumptions

The manifest is the numerical source of truth. The following table records its illustrative default input prices, not calibrated devices or literature measurements. A changed manifest requires fresh generated results, not manual repinning of prose results.

Table 5: Illustrative Energy Inputs

Input	Manifest value	Units	Evidence class
FLOP price	1.0	pJ/FLOP	Assumption
SRAM traffic price	0.10	pJ/byte	Assumption
DRAM traffic price	20.0	pJ/byte	Assumption
Spike-event price	1.0	aJ/spike	Assumption, not circuit measurement
Idle power	0.50	W	Assumption

Input	Manifest value	Units	Evidence class
Electrical actuator power	2.0	W/joint	Assumption
Sensor power	0.005	W/channel	Assumption

Landauer's $k_B T \ln 2$ (Landauer 1961) is a separate physical reference per erased bit at specified temperature, not one of these device prices. At 295 K it is approximately 2.82×10^{-21} J/bit. The transport convention uses $g = 9.81\,\mathrm{m/s^2}$ as a nominal gravitational acceleration, not an observed platform property. A mass of 0.001 kg is 1 g, not 1 mg.

13.2 B. Proposed Measurement Protocol

The following are requirements for a future calibration campaign, not procedures performed by the native null-meter example:

- Identify devices, measurement rails, instruments, firmware, sampling resolution, counter ranges/wrap rules, synchronization error, and calibration certificates.
- Specify environmental targets before acquisition. For example, 20 ± 2 °C and $45 \pm 5\%$ relative humidity would be protocol choices, not recorded conditions in this release.
- Preserve timestamped raw traces and benchmark inputs; report software and hardware versions.
- Distinguish independent devices/sessions/animals from repeated deterministic calculations or repeated readings on one unit.
- Fit coefficient uncertainty and evaluate held-out workloads; report measurement failure/unavailable status rather than fabricated zeros.
- For locomotion, align energy, duration, distance, mass, and physical boundary. Do not infer CoT from an unrelated workload window.

13.3 C. Parameterization and Model Selection

The example's morphology, population, activity, and horizon ranges are illustrative engineering dimensions. They are not a species library, universal operating limits, or inferred physiological optima. A hexapod with 18 modeled joint degrees of freedom is a six-by-three design assumption.

Selected CE workloads are **base** Body/Brain/Mind. Supported solvers are `pgs`, `si`, `ldlt`; supported neural patterns are `random`, `structured`. Base Mind uses canonical B, H_p and sums exact tree nodes. Enhanced Mind is separate: finite `max_policies` (default 1000) bounds leaf policies uniformly; `None` requests exact leaf enumeration. H_p is in steps; seconds require an explicit step duration.

13.4 D. Reproducibility and Acceptance Requirements

- Preserve manifest, invocation, model family, environment, package revision, result identity, and generated-artifact provenance.
- Require a completed coherent run before projection into paper-local compatibility paths; do not reuse concatenation snapshots or retained PDFs as fresh proof.
- Reconcile per-window rows with per-decision tables using the configured update count and explicit baseline allocation.
- Keep analytical results, assumptions, literature data, actual measurements, and proposed protocols distinct in captions/tables.

- Bootstrap effort (1000 resamples) and deterministic repetition (100 repeats) do not establish independent sample size or guaranteed power.
- Rendering and mathematical regression checks are implementation acceptance, not external scientific validation.

13.5 E. Notation

Symbol	Description
J, C, S	Joint DOF, contacts, sensor channels
K, N_{KC}, ρ, H	Input channels, population, activity fraction, state dimension
B, H_p	Branching factor, policy horizon in steps
G, E	Proposed field cells and deposits per update
e_F, e_S, e_D	FLOP and SRAM/DRAM traffic prices in pJ
E_{spk}	Spike-event price in aJ/spike
P_0	Idle power in W
t, f	Window duration in s, update rate in Hz

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